

Can a single X-ray observation tell a black hole from a neutron star in a source never seen in training?

Leakage-safe, pre-registered tests with RXTE, MAXI and NICER
(the XRB-pilot project, versions v1–v14)

Technical report

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Chapter 1

Data sources and physical feature extraction

This chapter defines, once, every input quantity used anywhere in the project: where the raw data come from, how a raw file becomes a number, in which unit, by which function of the code, and what happens to non-positive or missing values. The phase chapters (Chapter 5 and ??????????) refer back here and only state differences. When the description in a repository report and the code differ, the code is followed and the difference is flagged with a `% CHECK` comment in the `LATEX` source.

1.1 Instruments, data modes and catalogues

Table 1.1 lists every data product that enters a model or a test. All raw files were obtained by scripts; every download is recorded with its URL, size in bytes and SHA256 in `logs/downloads.jsonl` (50 773 records, ??). Raw data are not in git (`data/raw/` is ignored, large sets live outside the repository under `/xrb-pilot-data/`).

1.2 Source labels

The class label is a property of the *source*: $y = 1$ for a black hole (BH) and $y = 0$ for a neutron star (NS). Only labels with direct evidence are used for training and evaluation:

- **BH**: a dynamical mass function (radial-velocity curve of the companion). Versions v1–v4 used the dynamically confirmed systems of BlackCAT (?); v6 added high-mass X-ray binaries from ?; from v7 on the list of ? (25 systems in their Table 1) is used.
- **NS**: type-I X-ray bursts (thermonuclear burning on a solid surface) from ? (v1–v6) or the MINBAR catalogue (?) (v7c onwards), or coherent accretion-powered pulsations (?) (v6 external sources, v7c pulsar class; ? for HMXB pulsars in v7c).
- **BH candidates** (X-ray properties only, no dynamical mass) are never used as labels. In v3c they appear only as prediction targets and in a label-sensitivity analysis.

The label rules of every version are summarised in Table 1.2; the realised source lists are given in the phase chapters.

Table 1.1: Instruments, data modes and catalogues used in the project. “Versions” gives where the product enters a model or test. Band and resolution are those actually used by the code (not the instrument limits). Code: the function that reads or processes the product.

Product	Versions	What is used	Code
RXTE/PCA Standard-2 Stan- dard Products (xp*_s2.pha, _b2.pha, .rsp)	v1–v8, v10	129-channel total and <code>pcabackest</code> back- ground spectra summed over the PCUs listed in the <code>ROWID</code> keywords; 5–25 keV (3–25 keV in v3b); response for the same PCU combi- nation	<code>spectra.py: read_ pha, read_rsp, net_spectrum; 04_ fetch_process.py</code>
RXTE/PCA Standard-1 (pca/FS46_*)	v2–v10	0.125 s light curves per PCU (<code>XeCntPcu0–4</code>), plus array rates <code>VpCnt</code> , <code>RemainingCnt</code> , <code>VLECnt</code> ; 128 s rows of 1024 bins	<code>spectra.std1_ rates; v5lib.py; v6lib.py; v7lib.py</code>
RXTE/PCA event mode (pca/SE*.evt.gz)	v8c	Events with <code>TIME</code> , <code>PCUID</code> ; modes <code>E_{n}us_{m}M_0_{k}s</code> ; binned at 2^{-12} s	<code>v8lib.py: read_ events, hf_ features</code>
RXTE/HEXTE clus- ter B Standard Prod- ucts (xh*_s1.pha, _b1.pha)	v4b3	Net 25–40 and 40–60 keV count rates	<code>19_v4b3_hexte.py</code>
RXTE MissionLongData catalogues; HEASARC TAP xtemaster	v1–v8	Observation lists: <code>OBSID</code> , <code>TIME</code> , <code>EXPOSURE</code> , <code>STD1RATE</code> , <code>RA</code> , <code>DEC</code> , <code>NPCU</code>	<code>03_select_ observations.py, 20_v4b2_ select.py, 41_ v8b_select.py</code>
MAXI/GSC standard products, 1-day light curves	v7c, v8d, v9c	Photon fluxes in 2–4, 4–10, 10–20 keV	<code>38_v7c2_ select.py</code>
de Beurs et al. (2022) processed data (GitHub, commit 488b785)	v7c1	Their 44 sources, ten 20% subsamples, fea- tures (<code>RelInt</code> , <code>SC</code> , <code>HC</code>)	<code>37_v7c1_repro.py</code>
NICER/XTI cleaned events (ni*_Ompu7_cl.evt.gz); HEASARC TAP nicermastr	v9–v13	Events with <code>TIME</code> , <code>PI</code> , <code>DET_ID</code> ; 2–10 keV; 1/256 s bins, 128 s segments	<code>v9lib.py, v10lib.py, v11lib.py</code>
NICER unfiltered events (_ufa), NICER CALDB 3C50 library	v12, v13	Input of HEASoft <code>nibackgen3C50</code> (3C50 background model)	<code>v12_bkg3c50.sh, 58_v12_bkg.py, 62_v13_bkg.py</code>
MINBAR DR1 burst, observation and source tables	v7–v9	Burst start times and durations; burst-source positions	<code>v7lib.py: load_ minbar, gti_utc</code>
HI4PI (via HEASoft <code>nh</code>); XSPEC <code>tbabs</code>	v8d	Galactic N_H and band transmissions	<code>v8d_heasoft.sh, 45_v8d_nh.py</code>
CDS Sesame (SIM- BAD) positions	v4b2–v9	Source positions for label and pointing matches	<code>20_v4b2_ select.py, 28_ v6_select.py, 38_ v7c2_select.py</code>

Table 1.2: Source labelling and eligibility rules by version. “Pointing” is the maximum separation between an observation’s pointing and the source position used. Eligibility always also requires catalogue exposure ≥ 1000 s (RXTE: and $\text{STD1RATE} \geq 5 \text{ c s}^{-1} \text{ PCU}^{-1}$; NICER: `nicermastr` exposure ≥ 1000 s).

Version	BH rule	NS rule	Pointing / epoch / minimum
v1	4 BlackCAT dynamical BHs with ≥ 30 eligible pointings	4 Galloway+2008 bursters	0.1° from catalogue median; gain epoch 5, MJD 51677–54094
v2, v3	every BlackCAT Table 4 dynamical BH with a MissionLongData catalogue	every Galloway+2008 appendix-A source with a catalogue	0.1° ; epoch 5 (MJD 51677–55931); ≥ 10 eligible
v3c	16 BlackCAT candidates (prediction only)	–	as v2
v4b2	BlackCAT dynamical BHs matched by SIMBAD position	Galloway+2008 Table 3 bursters matched by position	all gain epochs; ≥ 10
v6 (external)	Casares & Jonker (2014) HMXB BHs	Patruno & Watts (2021) AMXPs (+ 4 slow pulsars as stress set)	0.1° from SIMBAD; all epochs; ≥ 10
v7b2	Cyg X-1, LMC X-1, LMC X-3 (Marcel+2026)	–	0.1° from SIMBAD; epoch 5; ≥ 10
v7c	Marcel+2026 Table 1	MINBAR bursters without pulsations (NPNS); pulsars (PSR) secondary	MAXI source within 0.1° (Sesame; 0.3° fallback); ≥ 100 daily points
v8b	unused Marcel+2026 BHs with RXTE data	–	0.1° from SIMBAD; all epochs; ≥ 3
v9–v13	Marcel+2026 Table 1 (Sesame positions)	MINBAR bursters, excluding entries with another entry within 0.05°	target within 0.05° ; ≥ 5

1.3 Observation eligibility and time-stratified sampling

For every source the eligible pointings are sorted by time and divided into K consecutive groups of (as nearly as possible) equal size, using `numpy.array_split` ($K = 15$ for RXTE, $K = 8$ for NICER). Inside each group the candidates are put in a random order generated by a seeded generator; the first candidate that passes the post-download quality rules (Section 1.4.5) is accepted. This keeps the sample spread over the source’s history (several outbursts, several states) and makes the choice independent of the spectrum. If a source has fewer than K eligible pointings, some groups are empty and all pointings are used. Algorithm 1 gives the procedure; the seeds are listed in Table 1.3.

Algorithm 1 Time-stratified candidate ranking and acceptance (`03_select_observations.py` + `04_fetch_process.py`; NICER: `47_v9_select.py`, `55_v11_select.py`, `60_v13_select.py`)

Input: eligible pointings $\{(o_i, t_i)\}$ of one source, number of groups K , seed σ , maximum tries M

```

1: sort pointings by time  $t_i$ ; split the index list into  $K$  consecutive groups  $G_1, \dots, G_K$  (array_split)
2:  $\text{rng} \leftarrow \text{default\_rng}(\sigma)$ 
3: for  $b = 1, \dots, K$  do
4:    $\pi \leftarrow \text{rng.permutation}(G_b)$  ▷ rank within the group
5:   for  $r = 1, \dots, \min(M, |G_b|)$  do
6:     download and process  $o_{\pi_r}$ ;
7:     if all quality rules pass then accept  $o_{\pi_r}$ ; break
8:   end if
9: end for
10: end for
```

Table 1.3: Sampling parameters. “position” is the 1-based position of the source in the source list processed by the script. The global seed is `SEED = 42` (`config.py`).

Sample	groups K	seed σ	max. tries M
RXTE v1, v2, v3c, v7b2, v8b	15	<code>SEED + position</code>	6
RXTE v4b2 new sources	15	<code>SEED + 100 + i</code>	6
RXTE v4b1	one group per pointing (all eligible)	–	1
NICER v9	8	<code>SEED + position</code>	3
NICER v11	8	<code>SEED + 1000 + position</code>	3
NICER v13	8	<code>SEED + 2000 + position</code>	3

1.4 RXTE/PCA Standard-2 spectra

1.4.1 What the files contain

The Standard Products of every ObsID contain a total (source + background) spectrum `xp<obs>_s2.pha`, the pcabackest model background spectrum `xp<obs>_b2.pha` and the combined response `xp<obs>_rsp`. The header inspection of `05_two_source_check.py` (`reports/step2_fits_check.md`) established: `HDUCLAS2/3 = TOTAL/COUNT`, column `COUNTS` in counts, `POISSERR = F` with explicit `STAT_ERR`, `SYS_ERR = 0`, `BACKSCAL = AREASCAL = 1`, 129 channels, equal source and background exposures, and the PCUs summed listed in the `ROWIDn` keywords. The background statistical error is $\sqrt{\text{model counts}}$; the systematic uncertainty of the background

model is *not* included anywhere. Standard Products are *not* deadtime corrected. Figure 1.1 shows the two spectra used for this check.

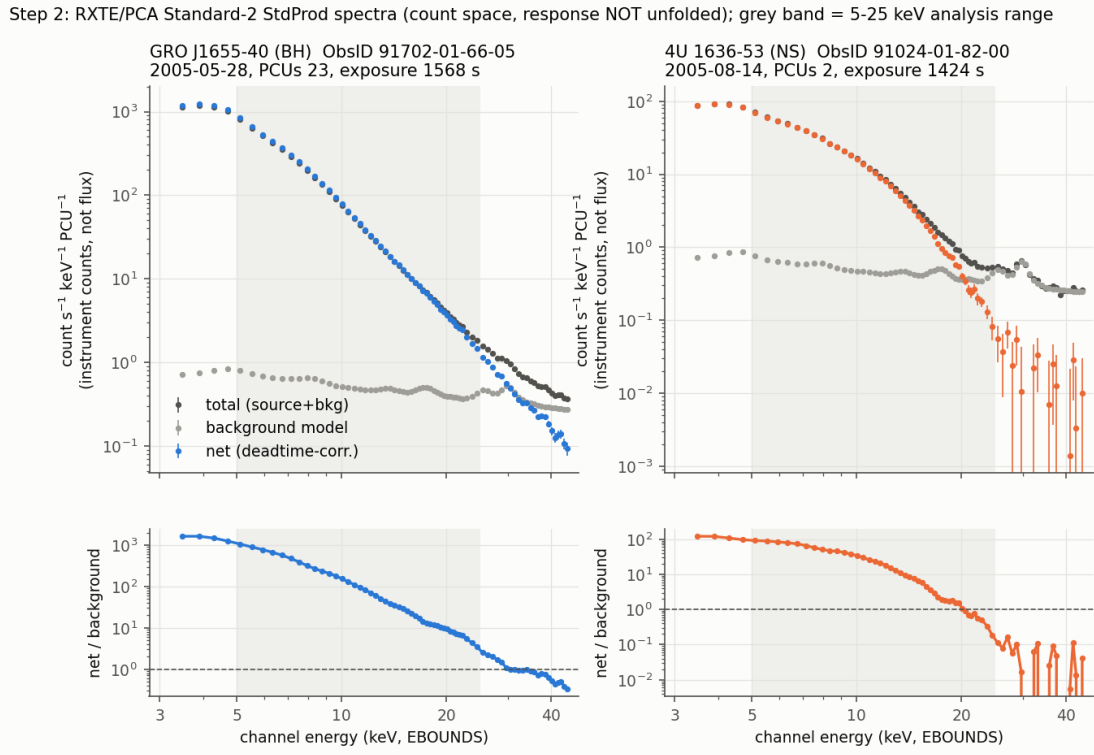


Figure 1.1: Data-flow validation on one BH (GRO J1655–40, ObsID 91702-01-66-05) and one NS (4U 1636–53, ObsID 91024-01-82-00). Top: total (dark grey), background model (light grey) and deadtime-corrected net (colour) count rates per keV per PCU against channel energy (EBOUNDS), log-log; the grey band is the 5–25 keV analysis range. Bottom: net/background ratio. The point is that the net spectrum is background-dominated only above ~ 30 keV for these bright sources, and that everything is in instrument count space (no response unfolding). Repo: `figures/step2_two_sources.png`, script `scripts/05_two_source_check.py`.

1.4.2 Background subtraction

For channel c of observation i , with source counts C_c^s and errors σ_c^s , background counts C_c^b and errors σ_c^b (function `spectra.net_spectrum`):

$$t_s = \text{EXPOSURE}_s \text{AREASCAL}_s, \quad t_b = \text{EXPOSURE}_b \text{AREASCAL}_b, \quad \kappa = \text{BACKSCAL}_s / \text{BACKSCAL}_b, \\ r_c^s = C_c^s / t_s, \quad e_c^s = \sigma_c^s / t_s, \quad r_c^b = \kappa C_c^b / t_b, \quad e_c^b = \kappa \sigma_c^b / t_b, \quad (1.1)$$

$$r_c^{\text{net}} = D (r_c^s - r_c^b), \quad e_c^{\text{net}} = D \sqrt{(e_c^s)^2 + (e_c^b)^2}, \quad (1.2)$$

in count s^{-1} (summed over the PCUs of the spectrum), where $D = 1/(1 - \text{DTF})$ is the deadtime correction factor (Section 1.4.3). This is the OGIP/XSPEC convention: the same quantity that XSPEC uses after `data`, `backgrnd` and `response`. Non-positive net rates (possible where the source is faint, mainly at $E > 20$ keV in soft spectra) are kept unchanged: no absolute value, no clipping, no deletion.

1.4.3 Deadtime

v1 (lower bound). Only the good-xenon term was available from Standard-2:

$$\text{DTF}^{(\text{v1})} = 10^{-5} \text{ s} \times \frac{\sum_c C_c^s}{t_s N_{\text{PCU}}}, \quad (1.3)$$

i.e. the total (source + background) count rate per PCU times $10 \mu\text{s}$. Median 0.31% and maximum 6% in the v1 sample (`data/observations.csv`, column `dtf_xe`).

v2 onwards (RXTE GOF recipe from Standard-1). The Standard-1 files of the ObsID are read and the mean rates inside the spectrum's good time intervals (GTIs) are computed from all 0.125 s bins whose centre lies in a GTI (`spectra.std1_rates`): per-PCU good-xenon rates X_{e_p} ($p = 0, \dots, 4$) and array rates V_p (propane), Rem (coincident events) and VLE (very large events). With N_{on} the number of PCUs with $X_{e_p} > 1 \text{ c s}^{-1}$ (`spectra.std1_dtf`):

$$\text{DTF}_p = 10^{-5} \left(X_{e_p} + \frac{V_p + \text{Rem}}{N_{\text{on}}} \right) + 6 \times 10^{-5} \frac{\text{VLE}}{N_{\text{on}}}, \quad \text{DTF} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \text{DTF}_p, \quad D = \frac{1}{1 - \text{DTF}}, \quad (1.4)$$

with rates in count s^{-1} and $\mathcal{P}$ the PCUs of the spectrum. A version with $1.5 \times 10^{-4} \text{ s}$ per VLE event (quoted elsewhere on the same GOF page) was stored only as a sensitivity number (`dtf_std1_vle150us`). In the v2 sample the Standard-1 coverage of the spectrum GTIs has median 99% (minimum 88%), the DTF has median 1.38% and maximum 8.8%; no observation exceeded the 10% rule. Because D multiplies the whole spectrum, it changes intensities (representation A) by 1–9% and leaves shapes (B, H) unchanged.

1.4.4 Common energy grid and rebinning

Gain drifts change the channel-to-energy relation slightly between observations, so every spectrum is rebinned onto one fixed grid defined by a reference observation (`REFERENCE_OBS = 91702-01-66-05`). Let $\{E_k^{\text{lo}}, E_k^{\text{hi}}\}$ be the EBOUNDS of the reference response. The grid edges are the union of all EBOUNDS edges between the edge closest to 5 keV and the edge closest to 25 keV: $e_0 = 4.908, \dots, e_{45} = 25.008 \text{ keV}$ (45 bins, widths 0.41–0.86 keV; Table 1.4). For v3b the lower limit is the edge closest to 3 keV, giving 50 bins from 2.883 keV; the 45 bins above 4.908 keV are identical. For an observation with its own EBOUNDS, the overlap matrix (`spectra.overlap_matrix`)

$$W_{jk} = \frac{\max(0, \min(E_k^{\text{hi}}, e_{j+1}) - \max(E_k^{\text{lo}}, e_j))}{E_k^{\text{hi}} - E_k^{\text{lo}}} \quad (1.5)$$

is the fraction of channel k that lies in grid bin j (counts assumed uniform within a channel). The rebinned net rate per keV per PCU and its error are

$$x_{ij} = \frac{\sum_k W_{jk} r_{ik}^{\text{net}}}{N_{\text{PCU},i} \Delta e_j}, \quad \sigma_{ij} = \frac{\sqrt{\sum_k W_{jk}^2 (e_{ik}^{\text{net}})^2}}{N_{\text{PCU},i} \Delta e_j}, \quad \Delta e_j = e_{j+1} - e_j, \quad (1.6)$$

in $\text{count s}^{-1} \text{ keV}^{-1} \text{ PCU}^{-1}$ (array `rate` and `err` of `features.npz`). Division by the number of PCUs is justified because the effective area grows linearly with the number of PCUs ($\approx 1070 \text{ cm}^2$ per PCU at 6 keV). The error formula treats channels as independent; two grid bins that share one channel are therefore weakly correlated, which is ignored. The total (source + background) and background spectra are rebinned in the same way (`srates`, `brate`). The gain drift within epoch 5 is at most 0.85 channel (median 0.17) relative to the reference.

Table 1.4: The 45-bin common energy grid (keV), from the EBOUNDS of observation 91702-01-66-05 (`data/v2/processed/features.npz`, key `edges`; identical in every RXTE sample). Bin j spans $[e_j, e_{j+1})$. The v3b grid adds five bins below: 2.883, 3.287, 3.691, 4.096, 4.502, 4.908 keV.

j	e_j	j	e_j	j	e_j	j	e_j	j	e_j
0	4.908	10	9.001	20	13.149	30	17.346	40	21.588
1	5.315	11	9.414	21	13.566	31	17.768	41	22.014
2	5.722	12	9.827	22	13.985	32	18.191	42	22.441
3	6.130	13	10.240	23	14.403	33	18.614	43	23.295
4	6.538	14	10.654	24	14.822	34	19.038	44	24.151
5	6.948	15	11.069	25	15.242	35	19.462	45	25.008
6	7.357	16	11.484	26	15.662	36	19.886		
7	7.767	17	11.899	27	16.082	37	20.311		
8	8.178	18	12.315	28	16.503	38	20.736		
9	8.589	19	12.732	29	16.924	39	21.162		

1.4.5 Summary quantities, errors and quality rules

For each observation (`04_fetch_process.py`):

$$F_i = \sum_{j=0}^{44} x_{ij} \Delta e_j \quad [\text{count s}^{-1} \text{PCU}^{-1}], \quad \sigma_{F,i} = \left(\sum_j (\sigma_{ij} \Delta e_j)^2 \right)^{1/2}, \quad S/N_i = F_i / \sigma_{F,i}, \quad (1.7)$$

$$f_i^{\text{bkg}} = \frac{\sum_j x_{ij}^b \Delta e_j}{\sum_j x_{ij}^s \Delta e_j} \quad (\text{background fraction, 5–25 keV}).$$

An observation is accepted only if all of the following hold (otherwise the next candidate of the same time group is tried): all files present and readable, 129 channels, EBOUNDS cover 5–25 keV; spectrum exposure ≥ 1000 s; $S/N \geq 10$; $DTF \leq 0.10$; header position `RA_OBJ/DEC_OBJ` within 0.1° of the source position; all values finite. $F_i > 0$ is guaranteed by the S/N rule, so the shape normalisation (Eq. (1.10)) is always defined.

1.4.6 Response check and response-normalised spectra

A fixed photon spectrum $N(E) = E^{-2}$ (photon index $\Gamma = 2$, no absorption, norm $1 \text{ ph cm}^{-2} \text{ s}^{-1} \text{ keV}^{-1}$ at 1 keV) is folded through each observation’s response matrix $R(E, c)$ (`spectra.powerlaw_folded`) and rebinned like the data:

$$P_{ic} = \sum_{\ell} R_i(E_{\ell}, c) \int_{E_{\ell}^{\text{lo}}}^{E_{\ell}^{\text{hi}}} E^{-2} dE, \quad p_{ij} = \frac{\sum_c W_{jc} P_{ic}}{N_{\text{PCU},i} \Delta e_j} \quad (\text{key p12}). \quad (1.8)$$

The across-observation scatter of $p_{ij} / \text{median}_i p_{ij}$ measures how much the instrument response alone changes the count spectrum: median 3.7%, maximum 11–12.7%, driven by the PCU combination (Figure 1.2). BH and NS have similar PCU-combination mixtures (mean tilt 2.0% vs 2.4% in v1), so this is treated as noise rather than as a class shortcut. From v4b2 on, p_{ij} also defines the response-normalised representations R and H_R (Section 1.5), which can be compared across gain epochs.

1.4.7 Independent check with XSPEC

For 12 randomly chosen v2 observations (seed 42; 10 sources; net rates $5\text{--}950 \text{ cs}^{-1}$), HEASoft 6.37.1/XSPEC (`data, backgrnd, response, ignore **-5.0 25.0-**, tclout rate`) and

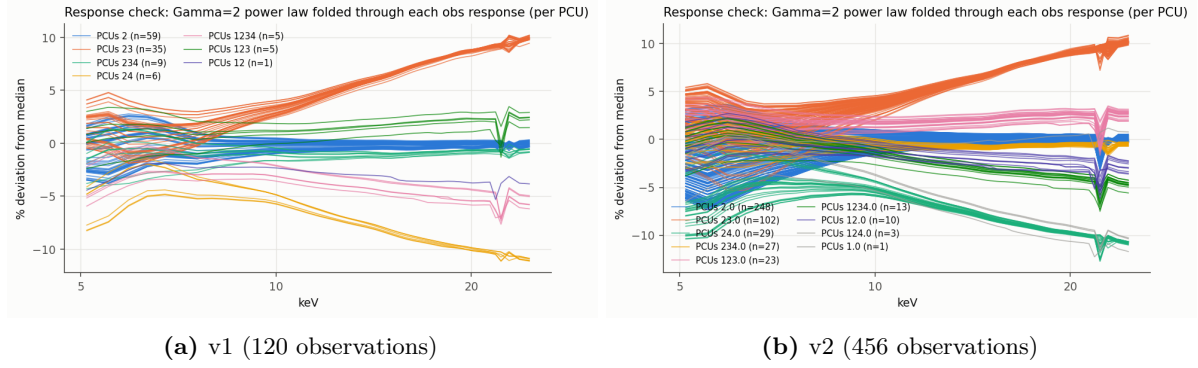


Figure 1.2: Response check: a fixed $\Gamma = 2$ power law folded through each observation’s response, per PCU, shown as percentage deviation from the median over observations, against energy; colour = PCU combination. The spread (a few per cent, up to $\sim 10\%$ at 25 keV) is set by which PCUs were on, not by the source class. Repo: `figures/step3_response_variation.png`, `figures/v2/step3_response_variation.png`; script `scripts/06_dataset_checks.py`.

Eq. (1.2) with $D = 1$ summed over the same noticed channels (10–52 or 10–53) agree to 3×10^{-10} (rate) and 4×10^{-10} (error) relative (`results/v2/xspec_crosscheck.csv`, `scripts/10_xspec_crosscheck.py`). This validates Eqs. (1.1) and (1.2); it does not validate the deadtime, the rebinning, or the background model itself.

1.5 Spectral representations

All representations are computed per observation from the rebinned spectrum x_i ; none uses information from other observations (the training-set standardisation of Section 3.1.1 is a separate step inside the model). Code: `07_train_eval.py`, `v3lib.v2_representations`, `12_v3b_extended_band.py`, `reps_R` in `22_v4b2_analysis.py`.

A (intensity kept). With scale $a = 0.1 \text{ count s}^{-1} \text{ keV}^{-1} \text{ PCU}^{-1}$ (`ASINH_SCALE`),

$$A_{ij} = \text{asinh}(x_{ij}/a), \quad j = 0, \dots, 44. \quad (1.9)$$

$\text{asinh}(u) \approx \text{sgn}(u) \ln(2|u|)$ for $|u| \gg 1$ and $\approx u$ near 0, so A is log-like for bright bins and defined for the (rare) negative bins. A keeps the absolute count rate, which depends on distance, absorption and accretion rate as well as on the compact object.

B (shape).

$$B_{ij} = x_{ij}/F_i \quad [\text{keV}^{-1}], \quad \sum_j B_{ij} \Delta e_j = 1. \quad (1.10)$$

H (two count-space colours). Four bands are formed from the grid bins whose geometric centre $\bar{e}_j = \sqrt{e_j e_{j+1}}$ lies in a nominal interval (`v3lib.band`); the exact bin ranges are in Table 1.5:

$$C_{i,m} = \sum_{j \in b_m} x_{ij} \Delta e_j \quad [\text{count s}^{-1} \text{ PCU}^{-1}], \quad \mathbf{h}_i = (h_{i1}, h_{i2}) = \left(\frac{C_{i,2}}{C_{i,1}}, \frac{C_{i,4}}{C_{i,3}} \right). \quad (1.11)$$

h_1 (“soft colour”, ≈ 7 –10 over 5–7 keV) and h_2 (“hard colour”, ≈ 16 –25 over 10–16 keV) are linear ratios. The pre-registration originally specified $\log_{10}$ colours; before any v2 model was trained, two BH spectra (95409-01-29-00 GX 339–4, 60113-01-34-00 XTE J1650–500) were

found with a slightly negative 16–25 keV net sum, so the linear ratio was adopted (logged in `reports/decision_log.md`). Denominators C_1, C_3 are positive for every observation (asserted in the code); numerators may be ≤ 0 and are kept.

HI, H+B, HI+A. $\text{HI}_i = (h_{i1}, h_{i2}, \log_{10} F_i)$; $\text{H+B}_i = (\mathbf{h}_i, B_{i0}, \dots, B_{i44})$ (47 features); $\text{HI+A}_i = (\text{HI}_i, A_{i0}, \dots, A_{i44})$ (48 features).

v3b (3–25 keV). On the 50-bin grid: $F_i^{(3)} = \sum_j x_{ij} \Delta e_j$ over all 50 bins; $\text{B3}_{ij} = x_{ij}/F_i^{(3)}$; $\text{H3}_i = (C_{i,0}/C_{i,1}, C_{i,2}/C_{i,1}, C_{i,4}/C_{i,3})$ where $C_{i,0}$ sums the five bins with centre in $[2.8, 5)$ keV (2.883–4.908 keV). The soft colour has 3–5 keV in the numerator and a denominator that is always positive.

R and H_R (response-normalised; v4b2 onwards). With p_{ij} of Eq. (1.8):

$$R_{ij} = \frac{x_{ij}}{p_{ij}}, \quad P_{i,m} = \sum_{j \in b_m} p_{ij} \Delta e_j, \quad \mathbf{h}_i^R = \left(\frac{C_{i,2}/P_{i,2}}{C_{i,1}/P_{i,1}}, \frac{C_{i,4}/P_{i,4}}{C_{i,3}/P_{i,3}} \right). \quad (1.12)$$

H_R is the colour of the data relative to the colour a $\Gamma = 2$ power law would have through the same response; it removes first-order response differences (e.g. between gain epochs). Its predictions agree with H at Spearman 0.994 (??).

Table 1.5: The four colour bands of H (and H_R): grid bins whose geometric centre lies in the nominal interval, and the actual energy range they cover (computed from `edges`).

Band	nominal interval (centre rule)	bins j	n bins	actual range [keV]
b_1	[5, 7)	0–4	5	4.908–6.948
b_2	[7, 10)	5–11	7	6.948–9.827
b_3	[10, 16)	12–26	15	9.827–16.082
b_4	[16, 25.1)	27–44	18	16.082–25.008
b_0 (v3b)	[2.8, 5)	5 bins of the 50-bin grid	5	2.883–4.908

1.6 HEXTE high-energy colours (v4b3)

Only HEXTE cluster B is used (`19_v4b3_hexte.py`), because it rocked between source and background until 2009-12-14 16:10 UT (MJD 55179.6736); observations on or after that date, or without cluster-B files, have no HEXTE features (60 of the 456 v2 observations). Header keyword `DEADAPP = T` (deadtime already applied) is required. The net cluster-B spectrum follows Eqs. (1.1) and (1.2) with $D = 1$; band rates $F_X(25\text{--}40)$ and $F_X(40\text{--}60)$ (count s^{−1} per cluster) use the overlap matrix of Eq. (1.5) on the RMF EBOUNDS. With $F_{\text{PCA}}(16\text{--}25) = C_{i,4}$ of Eq. (1.11):

$$X_1 = \frac{F_X(25\text{--}40)}{F_{\text{PCA}}(16\text{--}25)}, \quad X_2 = \frac{F_X(40\text{--}60)}{F_X(25\text{--}40)}. \quad (1.13)$$

X_1 is a cross-instrument count ratio and is only used as a feature. A colour is set to missing when its denominator has S/N < 3 (X_1 : 3 observations; X_2 : 144, of which 23 BH). Missing colours are filled with the training-fold median, and one 0/1 indicator per colour is appended (inside the LOSO loop; `hx_features`); H+X therefore has 6 columns.

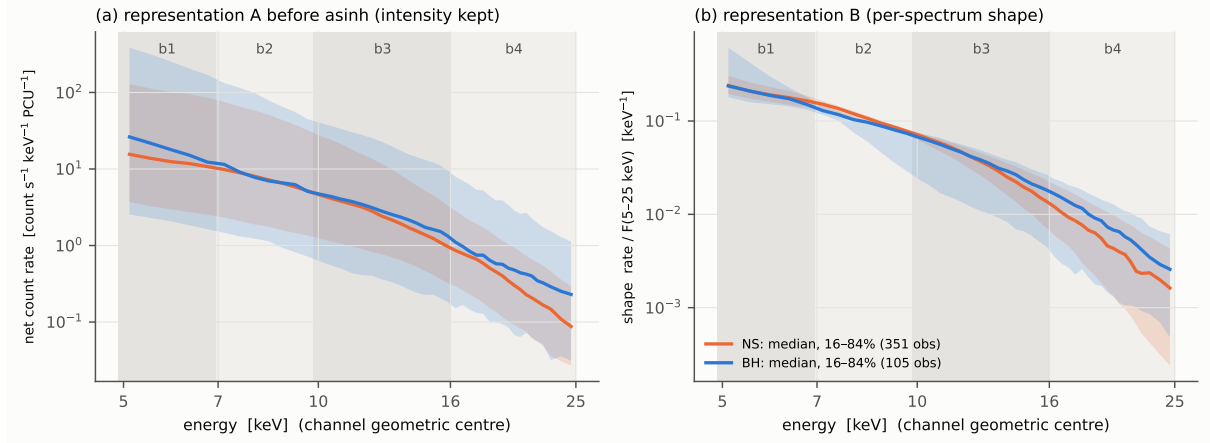


Figure 1.3: Median (line) and 16–84% range (shading) of the v2 count-space spectra, BH (blue, 105 observations) and NS (orange, 351 observations). (a) net rate per keV per PCU (the quantity transformed by asinh in representation A); (b) per-spectrum shape B. Grey and light bands mark the four colour bands b_1 – b_4 of Table 1.5. Non-positive bins are omitted from the percentiles. The class medians differ mainly in the overall slope above 10 keV and overlap strongly, which is why two colours capture most of the shape information (Chapter 5). Generated for this report from `data/v2/processed/features.npz` (`report_en/figures/generated/make_figures.py`).

1.7 Type-I X-ray bursts

Type-I bursts are seconds-long thermonuclear flashes that only NSs produce; an observation containing one carries a label shortcut and a distorted spectrum. Three procedures are used.

Standard-1 burst detector (v4; 17_v4_burst_detect.py). The 1 s light curve $r(t)$ is the sum of good-xenon counts of the spectrum’s PCUs over complete 1 s bins inside the GTIs. With μ, s the mean and standard deviation of all bins: candidates are bins with $r > \mu + 4s$; candidates closer than 30 s are merged into one event. For an event with peak r_{pk} at t_{pk} and persistent level r_0 (median of the 60 s before the event, or of the whole curve if fewer than 60 bins), amplitude $A = r_{\text{pk}} - r_0$ and levels $L_q = r_0 + qA$:

$$\text{burst} \iff r_{\text{pk}} \geq 1.5 r_0 \wedge t_{\text{rise}} \leq 10 \text{ s} \wedge 3 \leq t_{1/2} \leq 300 \text{ s} \wedge n_{25} \geq 3, \quad (1.14)$$

where $t_{\text{rise}} = t_{\text{pk}} - \max\{t < t_{\text{pk}} : r(t) < L_{0.25}\}$, $t_{1/2} = \min\{t > t_{\text{pk}} : r(t) < L_{0.5}\} - t_{\text{pk}}$, and n_{25} is the number of contiguous 1 s bins around the peak above $L_{0.25}$. The burst interval runs from the last time before the peak below $L_{0.25}$ to the first time after it below $L_{0.1}$. Thresholds are `V4_BURST` in `config.py`, following the search criterion of ?. Every candidate was also inspected by eye (`results/v4/burst_visual.csv`).

MINBAR matching (v7b1; v7lib.load_minbar, gti_utc, minbar_flag). The spectrum GTIs (STDGTI, mission elapsed time in TT) are converted to $\text{MJD}(\text{TT}) = \text{MJDREF} + t/86400$ and then to UTC with `astropy` ($\text{TT} - \text{UTC} = 64.18 \text{ s}$; round-trip error $< 1 \text{ ms}$). A MINBAR burst occupies $[T, T + \text{dur}]$ (dur missing $\Rightarrow 300 \text{ s}$). Primary rule: an observation is contaminated if a burst interval overlaps a GTI *and* the burst has the same ObsID or originates within 1° (PCA field of view) of the source. The literal pre-registered rule (any burst from any origin) is kept as a sensitivity set.

NICER burst rule (v9–v13; v9lib.burst_events). The same Eq. (1.14) is applied to the 2–10 keV 1 s light curve built from events inside the good intervals (Section 1.10); every 128 s

segment that overlaps $[t_{\text{start}} - 20 \text{ s}, t_{\text{end}} + 200 \text{ s}]$ of an automatically detected burst is removed.

1.8 Low-frequency timing from RXTE Standard-1 (v5–v10)

Figure 1.4 summarises the chain from light curve to features; this section gives the formulae.

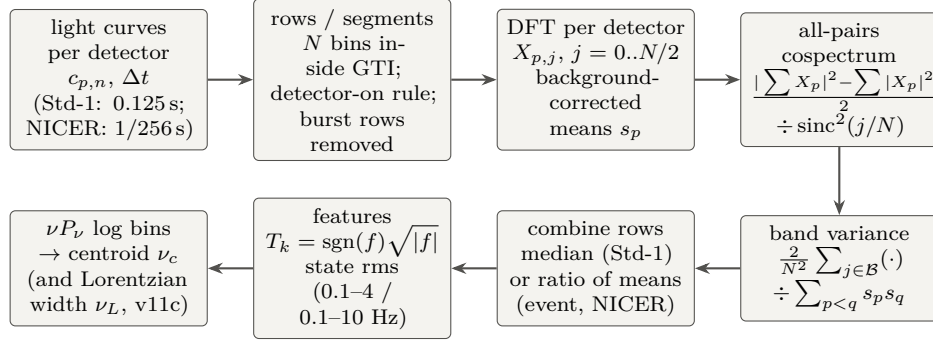


Figure 1.4: Timing-feature pipeline shared by RXTE Standard-1 (v5–v10), RXTE event mode (v8c) and NICER (v9–v13). Differences between the instruments are in the bin width, the segment length, the “detector” (PCU or NICER MPU), whether rows are combined by the median or by the ratio of means, and the frequency bands (Table 1.6). Drawn for this report.

1.8.1 Rows, detectors and background

A Standard-1 file stores, for each 128 s row, $N = 1024$ counts per PCU in $\Delta t = 0.125 \text{ s}$ bins. Only rows lying completely inside a spectrum GTI are used (start $\geq$ GTI start -1 ms and end $\leq$ GTI stop $+1 \text{ ms}$; `v5lib.std1_rows`). PCU p is *on* in a row if each of the eight 128-bin blocks (16 s) has at least one count (`v5lib.pcus_on`). The background count rate per PCU is taken from the Standard-2 background spectrum over all channels, $b = \sum_c C_c^b / (t_b N_{\text{PCU},b})$ counts s^{-1} (assumed equal for all PCUs). For a detector group or a single PCU with counts c_n and n_{det} detectors, the mean *source* counts per bin is $s = \bar{c} - n_{\text{det}} b \Delta t$; a row is invalid if any $s \leq 0$.

1.8.2 Fourier conventions and the binning correction

The discrete Fourier transform is $X_j = \sum_{n=0}^{N-1} c_n e^{-2\pi i j n / N}$, $j = 0, \dots, N/2$ (`numpy.fft.rfft`), frequency $\nu_j = j / (N \Delta t)$ (for 128 s rows, $\nu_j = j / 128 \text{ Hz}$). Parseval’s relation gives $\frac{1}{N} \sum_n (c_n - \bar{c})^2 = \frac{1}{N^2} \sum_{j \neq 0} |X_j|^2 \approx \frac{2}{N^2} \sum_{j=1}^{N/2} |X_j|^2$, so $\frac{2}{N^2} \sum_{j \in \mathcal{B}} |X_j|^2$ is the variance (in counts² per bin²) contributed by band $\mathcal{B}$. Integrating counts over bins of width Δt multiplies the Fourier amplitude by $\text{sinc}(j/N) = \sin(\pi j / N) / (\pi j / N)$ (? , Eq. 2.19); every power is divided by $\text{sinc}^2(j/N)$.

1.8.3 Two-group cospectrum (v5)

The PCUs that are on are sorted; those in even positions form group A, those in odd positions group B (at least two PCUs are required). With x_n, y_n the summed counts of the groups, X_j, Y_j their transforms, and s_x, s_y their background-corrected mean source counts per bin (`v5lib.row_features`):

$$V_{\mathcal{B}} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \frac{\text{Re}(X_j Y_j^*)}{\text{sinc}^2(j/N)}, \quad f_{\mathcal{B}} = \frac{V_{\mathcal{B}}}{s_x s_y}. \quad (1.15)$$

Because the Poisson (and deadtime) noise of different detectors is uncorrelated, the expected value of the cospectrum $\text{Re}(X_j Y_j^*)$ in the absence of source variability is zero (??); no Poisson level has to be modelled. $f_{\mathcal{B}}$ is the fractional variance (rms² normalised to the source mean) in band $\mathcal{B}$. Per observation, the *median* over valid rows is taken (robust to rare rows with flares or bursts); fewer than 3 valid rows $\Rightarrow$ missing. Features are signed square roots,

$$T_k = \text{sgn}(\text{median}_r f_{k,r}) \sqrt{|\text{median}_r f_{k,r}|}, \quad k \in \{1, 2, 3\}, \quad T_{\text{tot}} = \text{sgn}(F) \sqrt{|F|}, \quad F = \sum_k \text{median}_r f_{k,r}, \quad (1.16)$$

(fractional rms, which may be slightly negative because of noise). The bands are listed in Table 1.6. As an implementation check the conventional auto-power estimator $(|Z_j|^2 - N\bar{z})/\text{sinc}^2(j/N)$ of $z = x + y$ (Poisson level $N\bar{z}$, no deadtime) was also computed; for 276 S1 observations with 20–500 source $\text{cs}^{-1} \text{PCU}^{-1}$ the two versions of T_2 agree at Spearman 0.970.

1.8.4 All-pairs estimator and single-PCU hybrid (v6)

Using every pair of the n PCUs that are on, rather than one A/B split, reduces the noise (`v6lib.allpairs_row`). With per-PCU transforms $X_{p,j}$ and source means s_p :

$$\sum_{p < q} \text{Re}(X_{p,j} X_{q,j}^*) = \frac{|\sum_p X_{p,j}|^2 - \sum_p |X_{p,j}|^2}{2}, \quad f_{\mathcal{B}} = \frac{\frac{2}{N^2} \sum_{j \in \mathcal{B}} \frac{|\sum_p X_{p,j}|^2 - \sum_p |X_{p,j}|^2}{2 \text{sinc}^2(j/N)}}{\frac{(\sum_p s_p)^2 - \sum_p s_p^2}{2}}. \quad (1.17)$$

Rows with a single PCU on use the auto-power estimator $f_{\mathcal{B}} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \frac{|X_j|^2 - N\bar{c}}{\text{sinc}^2(j/N)} / s^2$, but only if the total count rate is $\leq 500 \text{cs}^{-1}$ (where the deadtime-free Poisson level is adequate); otherwise the row is invalid. T_k^* is the signed square root of the median over all valid rows (pair rows and single rows together). In synthetic tests the noise variance drops by 31%, 33% and 41% for 3, 4 and 5 PCUs (theory 33%, 33%, 40%) without bias; missing values over 8350 observations fall from 942 to 341.

1.8.5 Characteristic frequency ν_c (v6)

Nine logarithmic Fourier-index bins are formed from $\text{unique}(\text{round}(\text{geomspace}(1, 512, 10))) = \{1, 2, 4, 8, \dots, 512\}$ as $[1, 1], [2, 3], [4, 7], \dots, [256, 511]$ (Table 1.7). For each bin b , P_b is the median over rows of the fractional variance (Eq. (1.17)) in that bin; the bin centre is $\nu_b = \sqrt{j_{\text{lo}} j_{\text{hi}}}/128 \text{Hz}$ and its width in $\ln \nu$ is $\Delta_b = \ln((j_{\text{hi}} + 0.5)/(j_{\text{lo}} - 0.5))$. The νP_ν centroid is (`v6lib.nu_centroid`)

$$\ln \nu_c = \frac{\sum_b w_b \ln \nu_b}{\sum_b w_b}, \quad w_b = \frac{\max(P_b, 0)}{\Delta_b}, \quad (1.18)$$

missing if $T_{\text{tot}}^* < 0.05$ or $\sum_b w_b \leq 0$. Since P_b/Δ_b is the power per unit $\ln \nu$ (i.e. νP_ν), ν_c is the frequency around which most of the variability power per logarithmic frequency interval lies: a model-free proxy for a break or characteristic frequency. Synthetic sinusoids at 0.05, 0.5 and 3 Hz are recovered within one log bin.

1.8.6 State index (v7a)

The variability state of an observation is classified by its fractional rms in 0.1–4 Hz ($j = 13\text{--}511$), computed with Eq. (1.17) from rows with ≥ 2 PCUs only (no single-PCU rows, because their Poisson level would need a deadtime model), after removing every row that overlaps a burst

Table 1.6: Fourier-index bands and their frequencies. Standard-1 and NICER use 128 s rows or segments ($\nu_j = j/128$ Hz); RXTE event mode uses 16 s segments ($\nu_j = j/16$ Hz). From `config.py`.

Band	index j	frequency [Hz]	data	config name
T_1	2–12	0.0156–0.094	Std-1, NICER	V5_BANDS_J, V9_T_BANDS_J
T_2	13–128	0.102–1.000	Std-1, NICER	
T_3	129–511	1.008–3.992	Std-1, NICER	
state (v7a)	13–511	0.102–3.992	Std-1	V7_RMS_J
state (v9+)	13–1280	0.102–10.0	NICER	V9_STATE_J
white monitor	12288–16383	96–128	NICER	V9_WHITE_J
LC	16–63	1.0–3.94	event mode	V8_HF_BANDS_J
HF1	64–1023	4.0–63.9	event mode	
HF2	1024–8191	64–512	event mode	
HF3	8192–16383	512–1024	event mode	
NULL	24576–32767	1536–2048	event mode	

Table 1.7: Bins of the νP_ν centroid. Left: RXTE Standard-1 (v6; 9 bins, ν_b and Δ_b of Eq. (1.18)). Right: NICER (v9–v13; 13 octave bins $j \in [2^k, 2^{k+1} - 1]$, centre $2^{k+0.5}/128$ Hz). Computed from the code (`v6lib.py`, `v9lib.py`).

Standard-1 (v6)				NICER (v9+)		
$[j_{lo}, j_{hi}]$	ν_b [Hz]	range [Hz]	Δ_b	k	$[j_{lo}, j_{hi}]$	ν_k [Hz]
[1, 1]	0.0078	0.0039–0.0117	1.099	0	[1, 1]	0.011
[2, 3]	0.0191	0.0117–0.0273	0.847	1	[2, 3]	0.022
[4, 7]	0.0413	0.0273–0.0586	0.762	2	[4, 7]	0.044
[8, 15]	0.0856	0.0586–0.121	0.726	3	[8, 15]	0.088
[16, 31]	0.174	0.121–0.246	0.709	4	[16, 31]	0.177
[32, 63]	0.351	0.246–0.496	0.701	5	[32, 63]	0.354
[64, 127]	0.704	0.496–0.996	0.697	6	[64, 127]	0.707
[128, 255]	1.411	0.996–1.996	0.695	7	[128, 255]	1.414
[256, 511]	2.826	1.996–3.996	0.694	8	[256, 511]	2.828
				9–12	[512, 1023]–[4096, 8191]	5.66, 11.3, 22.6, 45.3

interval $[t_{\text{start}} - 20 \text{ s}, \text{end}]$ (MINBAR field-of-view rule and the v4 detector; `v7lib.state_rms`). With $\rho = \text{sgn}(\text{median}_r f) \sqrt{|\text{median}_r f|}$:

$$\text{state} = \begin{cases} \text{hard-like} & \rho > 0.10 \\ \text{soft-like} & \rho < 0.075 \\ \text{intermediate} & \text{otherwise} \end{cases} \quad (\text{unknown if fewer than 3 rows}), \quad (1.19)$$

the thresholds of ρ , Table 2, which are defined for 0.1–10 Hz; the 4 Hz Nyquist limit of Standard-1 makes ρ slightly low and the hard-like class conservative. NICER (v9 onwards) uses the full 0.1–10 Hz band. “Soft-like/hard-like” are operational variability classes, not full spectral-state assignments.

1.9 High-frequency timing from RXTE event mode (v8c)

An event file qualifies if `DATAMODE` matches `^E_\d+us_\d+M_0_\d+s$`, `TDDES2` contains the six anode chains `X1L^X1R^X2L^X2R^X3L^X3R`, `TIMEDEL` $\leq 2^{-12} \text{ s}$ and a `PCUID` column exists (`v8lib.mode_qualifies`); if several modes qualify, the one with the largest total size is used. Events are binned at $\Delta t = 2^{-12} \text{ s}$ (Nyquist 2048 Hz) in 16 s segments ($N = 65536$) lying completely in $(\text{spectrum GTI}) \cap (\text{event-file GTI})$ and not overlapping a burst interval. A PCU is on if each of the 16 one-second sub-intervals has an event; ≥ 2 PCUs are required. For segment r , with Eq. (1.17), $\text{num}_{\mathcal{B},r} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \text{cross}_j$ and $\text{den}_r = \sum_{p < q} s_p s_q$. The pre-registered per-observation estimator was the mean of the per-segment ratios; because single 16 s segments of faint sources give unstable ratios (one Aql X-1 segment gave -195), it was replaced, after the implementation check IC4 and before any model, by the ratio of means with a delta-method standard error (`v8lib.hf_summary`):

$$\hat{R}_{\mathcal{B}} = \frac{\sum_r \text{num}_{\mathcal{B},r}}{\sum_r \text{den}_r}, \quad \widehat{\text{SE}}(\hat{R}_{\mathcal{B}}) = \frac{1}{\sum_r \text{den}_r} \sqrt{\frac{n}{n-1} \sum_r (\text{num}_{\mathcal{B},r} - \hat{R}_{\mathcal{B}} \text{den}_r)^2}, \quad (1.20)$$

with n segments; features are $\text{sgn}(\hat{R}) \sqrt{|\hat{R}|}$ for LC, HF1, HF2, HF3. Quality: $n \geq 10$ and $|z_{\text{NULL}}| = |\hat{R}_{\text{NULL}} / \widehat{\text{SE}}| \leq 5$ in the empty 1536–2048 Hz band. A post hoc correction subtracts the null-band power scaled by bandwidth from every band $(\hat{R}_{\mathcal{B}} - \hat{R}_{\text{NULL}} |\mathcal{B}| / |\text{NULL}|)$, because a white cross-PCU correlated component was found (??).

1.10 NICER/XTI features (v9–v13)

1.10.1 Reading the events

Cleaned event files are read without HEASoft by a streaming FITS parser (`v9lib.PrefixParser`) that decompresses the gzip stream and extracts `TIME`, `PI` and `DET_ID` from the binary table rows. Three reading modes were used: v9 downloaded only the *prefix* of each file (HTTP range requests) until ≥ 6 gap-free 128 s segments, end of file, or 300 MB; v10 appended the *tail* up to 1 GB compressed and parsed prefix + tail as one stream; v11 and v13 *streamed* whole files (up to 1 GB) into memory without storing them. Parsing was checked to be identical to `astropy` and across the three modes (IC1 of v9–v11).

1.10.2 Good intervals, segments and detectors

Good intervals are maximal stretches without a gap $> 1 \text{ s}$ between consecutive events of any energy (`v9lib.good_intervals`). Each good interval $[a, b]$ is cut sequentially into 128 s segments

starting at a . Only 2–10 keV events are used for timing (PI 200–999, PI in units of 10 eV). The detector unit is the MPU, $m = \lfloor \text{DET_ID}/10 \rfloor \in \{0, \dots, 6\}$. Events are binned at $\Delta t = 1/256$ s ($N = 32768$, Nyquist 128 Hz). MPU m is on in a segment if each of the eight 16 s blocks has a 2–10 keV event; ≥ 2 MPUs are required. Segments overlapping a burst interval (Section 1.7) are removed.

1.10.3 Per-segment statistics and observation features

For each valid segment r (`v9lib.segment_stats`): s_m is the mean 2–10 keV counts per bin of MPU m (no background subtraction), the all-pairs cross-power is formed as in Eq. (1.17), and for each band $\text{num}_{\mathcal{B},r} = \frac{2}{N^2} \sum_{j \in \mathcal{B}} \text{cross}_j$, $\text{den}_r = \sum_{m < m'} s_m s_{m'}$. Observation features use the ratio of means (Eq. (1.20)) for T_1, T_2, T_3 , the state band 0.1–10 Hz and the 96–128 Hz white monitor, and the signed square root. In addition:

- **Colours** (`c1`, `c2`): counts of *all* events in the accepted segment windows in PI 200–399 (A , 2–4 keV), 400–599 (B , 4–6 keV) and 600–999 (C , 6–10 keV); $c_1 = B/A$, $c_2 = C/B$.
- **Count rate** `rate_2_10`: mean over segments of the 2–10 keV counts of the MPUs that are on, divided by 128 s.
- ν_c : 13 octave bins $k = 0, \dots, 12$ with $j \in [2^k, 2^{k+1} - 1]$ (1/128–64 Hz, Table 1.7); $\ell_k = \sum_r \text{num}_{k,r} / \sum_r \text{den}_r$; $\nu_c = \exp(\sum_k w_k \ln \nu_k / \sum_k w_k)$ with $w_k = \max(\ell_k, 0)$ and $\nu_k = 2^{k+0.5}/128$ Hz; missing if $\sqrt{\max(\sum_k \ell_k, 0)} < 0.05$.
- **State**: Eq. (1.19) applied to the 0.1–10 Hz rms.
- **Quality**: at least 3 valid segments and `rate_2_10` $\geq 20 \text{ cs}^{-1}$ (background then $\lesssim 5\%$ by the pre-registered estimate).

1.10.4 Background proxy (v10)

$\beta_i = N(\text{PI } 1200\text{--}1499)/N(\text{PI } 200\text{--}999)$, the 12–15 keV to 2–10 keV event ratio of the whole file (`v10lib.bkg_ratio`); the source contributes little above 12 keV, so a high β indicates background. The threshold is the 99th percentile of β among observations with `rate_2_10` $\geq 500 \text{ cs}^{-1}$ (where background is negligible); it was frozen at 0.2687 and reused in v11.

1.10.5 3C50 background correction and screening (v12, v13)

For observations with `rate_2_10` $< 500 \text{ cs}^{-1}$, HEASoft `nibackgen3C50` (?) was run inside WSL with default parameters and the NICER CALDB (`scripts/v12_bkg3c50.sh`); it returns a total spectrum $T(\text{PI})$ and a background spectrum $B(\text{PI})$. Band background fractions are $f_{\mathcal{X}} = \sum_{\mathcal{X}} B / \sum_{\mathcal{X}} T$ for $\mathcal{X} \in \{A, B, C, T = 200\text{--}999\}$. Brighter observations get $f = 0$. The frozen correction (`corrected` in `59_v12_analysis.py`, copied verbatim into `63_v13_analysis.py`) is

$$T'_k = \frac{T_k}{1 - f_T} \quad (k = 1, 2, 3, \text{state}), \quad c'_1 = \frac{B(1 - f_B)}{A(1 - f_A)}, \quad c'_2 = \frac{C(1 - f_C)}{B(1 - f_B)}, \quad (1.21)$$

followed by re-classification of the state; ν_c is not changed (it depends only on the shape of the power spectrum). The first relation holds because a fractional rms is normalised to the total mean; normalising to the source mean $S = (1 - f)(S + B)$ multiplies it by $1/(1 - f)$ (assuming a constant background). Screening keeps an observation if it is bright, or if 3C50 succeeded and $f_T \leq 0.20$. 3C50 failures (mostly data processed with the oldest gain calibration, which the tool rejects) are excluded. In v13 a descriptive variant recomputed the faint observations'

features from events inside the GTIs of the 3C50 total spectrum only (`gti_restricted` in `63_v13_analysis.py`).

1.10.6 Lorentzian fits (v11c)

The 13 octave-bin fractional variances ℓ_k (with delta-method errors, `v11lib.lb_se`) are fitted by one or two zero-centred Lorentzians whose integrated variance in a bin $[\nu_k^{\text{lo}}, \nu_k^{\text{hi}}]$ (bin edges $(j \mp 0.5)/128$ Hz) is

$$L_k(r^2, \Delta) = r^2 \frac{2}{\pi} \left[\arctan \frac{\nu_k^{\text{hi}}}{\Delta} - \arctan \frac{\nu_k^{\text{lo}}}{\Delta} \right], \quad \Delta \in [1/128, 128] \text{ Hz}, \quad (1.22)$$

by weighted least squares in log-parameters (`scipy.optimize.least_squares`, four starting points each, `v11lib.fit_lorentz`); $\text{BIC} = \chi^2 + k \ln n$ with $k = 2$ or 4 parameters; two components are preferred if $\text{BIC}_1 - \text{BIC}_2 > 6$. The width of the lowest-frequency component, ν_L , is the “break” frequency.

1.11 MAXI/GSC features (v7c, v8d, v9c)

The MAXI 1-day standard light curve of a source gives photon fluxes L (2–4 keV), M (4–10 keV), H (10–20 keV) and errors. Points are kept if every band has flux/error $\geq \sqrt{3}$ and positive errors, then points with $L + M + H \geq (\bar{L} + \bar{M} + \bar{H}) + 10 \text{ sd}$ are dropped; sources need ≥ 100 points (cuts of ?). Features per point (`39_v7c2_analysis.py`):

$$\text{SC} = \frac{M - L}{M + L}, \quad \text{HC} = \frac{H - L}{H + L}, \quad C_4 = \frac{H - M}{H + M}, \quad \text{RelInt} = \frac{L + M + H}{Q_{99.99}(L + M + H)}, \quad (1.23)$$

where $Q_{99.99}$ is the 99.99th percentile over that source’s points. Feature sets: $\text{CCI} = (\text{SC}, \text{HC}, \text{RelInt})$, $\text{C2D} = (\text{SC}, \text{HC})$, $\text{C1D} = (C_4)$ (the only colour without 2–4 keV).

N_{H} correction (v8d; `45_v8d_nh.py`, `v8d_heasoft.sh`). The Galactic HI column N_{H} at each source position is the HI4PI (?) weighted average within 0.1° (HEASoft `nh`, output `avwnh`). XSPEC computes photon fluxes of `tbabs*powerlaw` ($\Gamma = 2$; `abund wilm`, `xsect vern`; ?) on a grid $N_{\text{H}} \in \{0\} \cup \{10^{19}, \dots, 10^{23.5}\} \text{ cm}^{-2}$ (200 log steps); the transmission is $t_{\mathcal{X}}(N_{\text{H}}) = \Phi_{\mathcal{X}}(N_{\text{H}})/\Phi_{\mathcal{X}}(0)$ (e.g. 0.864, 0.979, 0.997 in L, M, H at 10^{22} cm^{-2}), interpolated linearly in $\log N_{\text{H}}$. Corrected fluxes $L' = L/t_L$, $M' = M/t_M$, $H' = H/t_H$ give SC' , HC' , C_4' , RelInt' by Eq. (1.23) (sets C2D_NH , C1D_NH , CCI_NH).

1.12 Feature summary

Table 1.8: Summary of every feature that enters a model or test. Missing-value rule: “excluded” = the observation does not enter; “median+ind.” = training-fold median and 0/1 indicator (Section 2.2.1); “term = 0” = CCTLR timing term set to zero.

Feature	Unit	Definition	Code	Non-positive / missing
x_{ij} (rate)	$\text{cs}^{-1} \text{keV}^{-1} \text{keV}$	$\text{RQU}(1.6)$	<code>04_fetch_process.py</code>	kept as is

Feature	Unit	Definition	Code	Non-positive / missing
A_{ij}	–	Eq. (1.9)	v2_representations	asinh defined for < 0
B_{ij}	keV $^{-1}$	Eq. (1.10)	same	$F > 0$ guaranteed
h_1, h_2	–	Eq. (1.11)	same	denominators > 0 ; numerators kept
$\log_{10} F$	dex	Eq. (1.7)	same	$F > 0$
B3, H3	–	Section 1.5	12_v3b_extended_band.py	as above
$R_{ij}, h_{1,2}^R$	–	Eq. (1.12)	reps_R	as above
X_1, X_2	–	Eq. (1.13)	19_v4b3_hexte.py	S/N $< 3 \rightarrow$ median+ind. (per colour)
T_1, T_2, T_3 (v5)	frac. rms	Eqs. (1.15) and (1.16)	v5lib.obs_timing	< 3 rows $\rightarrow$ median+ind.
T_k^* (v6)	frac. rms	Eq. (1.17)	v6lib.obs_timing_v6	median+ind.; CCTLR: term = 0
ν_c (RXTE)	Hz	Eq. (1.18)	v6lib.nu_centroid	missing $\rightarrow$ observation not in test
state rms (v7a)	frac. rms	Eq. (1.19)	v7lib.state_rms	< 3 rows $\rightarrow$ “unknown”
LC, HF1–3	frac. rms	Eq. (1.20)	v8lib.hf_features	QC fail $\rightarrow$ excluded
c_1, c_2 (NICER)	–	Section 1.10	v9lib.obs_features	excluded if QC fails
T_k , state (NICER)	frac. rms	Eq. (1.20)	same	excluded if QC fails
ν_c (NICER)	Hz	Section 1.10	same	missing $\rightarrow$ not in test
β	–	Section 1.10	v10lib.bkg_ratio	–
f_A, f_B, f_C, f_T	–	Eq. (1.21)	58_v12_bkg.py	3C50 failure $\rightarrow$ excluded
ν_L	Hz	Eq. (1.22)	v11lib.fit_lorentz	fit failure $\rightarrow$ missing
SC, HC, C_4 , RelInt	–	Eq. (1.23)	39_v7c2_analysis.py	cuts in Section 1.11
SC', HC', C'_4	–	Section 1.11	45_v8d_nh.py	–

Chapter 2

Data model and machine-learning interface

This chapter fixes the objects that every analysis passes to a model, so that each analysis in the phase chapters can be re-implemented from its array table alone. The per-version tables (actual $n \times p$, column meanings, source files) are given in each phase chapter under “Data structure in this phase”; this chapter defines the common notation, the generic recipe, the catalogue of stored arrays, and the leakage controls.

2.1 Notation

An analysis uses n observations of S sources.

- $\mathbf{X} \in \mathbb{R}^{n \times p}$: **design matrix**; row i is one observation (one RXTE ObsID, one NICER ObsID, or one MAXI 1-day point), column ℓ one feature of Chapter 1.
- $\mathbf{y} \in \{0, 1\}^n$: labels, $y_i = 1$ for BH, inherited from the source ($y_i = Y_{g_i}$).
- $\mathbf{g} \in \{1, \dots, S\}^n$: source (group) index of each row. $n_k = |\{i : g_i = k\}|$; $\mathcal{S}_{\text{BH}}$ and $\mathcal{S}_{\text{NS}}$ are the BH and NS source sets, $S_1 = |\mathcal{S}_{\text{BH}}|$, $S_0 = |\mathcal{S}_{\text{NS}}|$.
- $\mathbf{o}$: observation identifiers (`obs_id`; `obsid` for NICER); all joins between tables are by this key.
- $\mathbf{w} \in \mathbb{R}_{>0}^n$: sample weights used in a fit (class-balanced or source-equal; Section 3.1.2).
- $\phi : \{1, \dots, n\} \rightarrow \{1, \dots, S\}$: **fold assignment**; in leave-one-source-out (LOSO) $\phi(i) = g_i$, so fold k tests exactly the observations of source k .
- $\hat{\mathbf{s}} \in [0, 1]^n$: **out-of-fold (OOF) score**; $\hat{s}_i$ is produced by a model that was fitted without any observation of source g_i . Scores are model outputs on $[0, 1]$ (“BH score”), not calibrated probabilities.

Source-level score and decision. For source k ,

$$\bar{s}_k = \frac{1}{n_k} \sum_{i: g_i=k} \hat{s}_i, \quad \hat{Y}_k = \mathbf{1}[\bar{s}_k \geq t], \quad t = 0.5, \quad (2.1)$$

and the observation-level decision is $\hat{y}_i = \mathbf{1}[\hat{s}_i \geq t]$. The threshold t is fixed at 0.5 in every analysis except the nested procedures of v4a, where a fold-specific threshold t_k is selected inside the training sources and the stored score is shifted to $\hat{s}_i = \hat{s}_i^{\text{raw}} - t_k + 0.5$ so that the fixed 0.5

rule reproduces it (Section 4.2). Observations of one source are never treated as independent evidence: there is no product of probabilities over observations.

2.2 From files to matrices

Algorithm 2 is the generic recipe used by every analysis script. Two properties are essential: (i) every transformation that uses information from more than one observation (standardisation, imputation, residualisation, threshold choice) is computed *inside* the training part of a fold; (ii) rows are matched by identifier, never by position, except where the code asserts that the orders are identical.

Algorithm 2 Building $(\mathbf{X}, \mathbf{y}, \mathbf{g})$ for one analysis (generic; concrete versions in the phase chapters)

Input: spectral archive `data/<v>/processed/features.npz`; optional feature tables keyed by `obs_id`

- 1: load `rate`, `edges`, `F`, `pl2`, `y`, `source_id`, `obs_id` (one row per accepted observation)
- 2: compute the per-observation representation(s) of Section 1.5: `A`, `B`, `H`, `HI`, `R`, `HR` $\triangleright$ no cross-row information
- 3: optionally concatenate several samples (e.g. `v2 + v4b2`) by stacking rows; assert identical `edges`
- 4: for timing / HEXTE / state columns: `table.reindex(obs_id)`; non-finite $\rightarrow$ NaN
- 5: $\mathbf{X} \leftarrow$ column-stack of the chosen blocks; $\mathbf{y} \leftarrow \mathbf{y}$; $\mathbf{g} \leftarrow \mathbf{source_id}$
- 6: `folds  $\leftarrow$  LeaveOneGroupOut().split( $\mathbf{X}, \mathbf{y}, \mathbf{g}$ )`; assert one test source per fold, disjoint sources, both classes in training
- 7: **for** each fold $(\mathcal{T}, \mathcal{V})$ **do**
- 8: fill missing values with training medians + indicator (Eq. (2.2)), using $\mathbf{X}_{\mathcal{T}}$ only
- 9: fit the model (scaler inside the model, weights from $\mathbf{y}_{\mathcal{T}}, \mathbf{g}_{\mathcal{T}}$) and score $\mathbf{X}_{\mathcal{V}}$
- 10: **end for**
- 11: write the OOF table (`representation`, `model`, `fold`, `source_id`, `obs_id`, `true_label`, `BH_score`, `predicted_label`)

2.2.1 Missing values

Spectral features are never missing (observations failing the quality rules are not in the sample). Timing features can be missing (fewer than 3 valid rows, e.g. only one PCU on). For a block $\mathbf{T} \in (\mathbb{R} \cup \{\text{NaN}\})^{n \times q}$ and a fold with training rows $\mathcal{T}$ (`v5lib.fill_missing`):

$$m_{\ell} = \text{median}_{i \in \mathcal{T}, T_{i\ell} \text{ finite}} T_{i\ell} \text{ (0 if none)}, \quad \tilde{T}_{i\ell} = \begin{cases} T_{i\ell} & \text{finite} \\ m_{\ell} & \text{otherwise} \end{cases}, \quad \mu_i = \mathbf{1}[\exists \ell : T_{i\ell} \text{ non-finite}], \quad (2.2)$$

and $[\tilde{\mathbf{T}}, \boldsymbol{\mu}]$ (one indicator column) replaces $\mathbf{T}$ for both training and test rows. HEXTE colours use the same rule with one indicator per colour (Section 1.6). The CCTLR classifier needs no imputation: its timing term is set to zero for a row with any missing timing value (Section 3.11). NICER features are never imputed; observations failing the NICER quality rules are excluded. For the frozen external tests (v6a, v8b), the medians are taken from the S1 training set and applied unchanged to the external rows.

2.3 Catalogue of stored arrays

Table 2.1 lists every stored array from which a design matrix is built. Shapes and dtypes were read from the committed files with Python for this report.

Table 2.1: Catalogue of stored arrays (committed files). Shapes are the actual shapes in the repository. “c/s/keV/PCU” = count s⁻¹ keV⁻¹ PCU⁻¹. Every `features.npz` has the same keys: `rate`, `err`, `srates`, `brate`, `p12` ($n \times 45$, float64, c/s/keV/PCU: net rate, its error, total rate, background rate, folded $\Gamma = 2$ power law), `edges` (46, float64, keV), `obs_id` (n , unicode), `source_id` (n , unicode = *g*), `y` (n , int64 = *y*), `F` (n , float64, c/s/PCU).

File	key / columns	shape	content
<code>data/processed/features.npz</code>	all keys	120×45	v1: 8 sources (4 BH), 60 BH rows
<code>data/v2/processed/features.npz</code>	all keys	456×45	v2 = S1: 31 sources (7 BH), 105 BH rows
<code>data/v3/processed/features_3_25.npz</code>	<code>rate</code> , <code>err</code> , <code>edges</code> , <code>ids</code> , <code>y</code>	456×50	v3b: 2.883–25.008 keV grid
<code>data/v3c/processed/features.npz</code>	all keys	210×45	14 BH candidates ($y=1$, never used as label in evaluation)
<code>data/v4b1/processed/features.npz</code>	all keys	7949×45	S3: every eligible epoch-5 pointing of the 31 sources (2369 BH rows)
<code>data/v4b2/processed/features.npz</code>	all keys	221×45	15 new sources (2 BH), all gain epochs
<code>data/v6/processed/features.npz</code>	all keys	180×45	12 external sources (2 BH; 4 are slow-pulsar stress sources)
<code>data/v7b1/processed/features_replaced.npz</code>	all keys	435×45	v2 with burst-contaminated observations replaced
<code>data/v7b2/processed/features.npz</code>	all keys	45×45	Cyg X-1, LMC X-1, LMC X-3
<code>data/v8b/processed/features.npz</code>	all keys	23×45	GS 1354–64, SS 433
<code>data/v4b3/processed/hexte_features.csv</code>	<code>X1</code> , <code>X2</code> , <code>status</code> , <code>errors</code> , <code>S/N</code>	456 rows	HEXTE colours of the v2 observations
<code>data/v5/processed/timing_features.csv</code>	<code>T1</code> , <code>T2</code> , <code>T3</code> , <code>T_total</code> , <code>status</code> , <code>in_v2/v4b2/v4b1</code>	8170 rows	v5 two-group estimator for $S1 \cup S2 \cup S3$
<code>data/v6/processed/timing_v6.csv</code>	<code>T1s</code> , <code>T2s</code> , <code>T3s</code> , <code>T_total</code> , <code>nu_c</code> , <code>lb0..lb8</code> , <code>split halves</code>	8350 rows	v6 all-pairs estimator and ν_c
<code>data/v6/processed/timing_v5_new.csv</code>	as v5	180 rows	v5 estimator for the v6 sources
<code>results/v7a/states.csv</code>	<code>rms</code> , <code>state</code> , <code>c1</code> , <code>c2</code>	456 rows	v7a state of every S1 observation
<code>data/v8b/processed/external_states_timing.csv</code>	<code>rate</code> , <code>T1..T3</code> , <code>role</code>	439 rows	state and timing of external set X (incl. stress sources)

File	key / columns	shape	content
data/v8c/processed/hf_features.csv	LC, HF1..HF3, NULL, *_var, *_se, *_z, status	311 rows	event-mode features (148 S1 rows pass QC)
data/v10/processed/rxte_timing_table.csv	c1R,c2R, state, nu_c, T1s..T3s	895 rows	RXTE hard-state ν_c table for v10b
data/v7c/maxi_points.csv	mjd, L,M,H (+errors), source, cls	132 057 rows	MAXI daily points after cuts (BH, NPNS, PSR)
data/v8d/nh_hi4pi.csv; tbabs_transmission.csv	avwnh; t_L, t_M, t_H	100; 201 rows	HI4PI N_H ; transmission table
data/v9/processed/nicer_features.csv	c1,c2,T1..T3,STATE,nu_c,lb0..lb12,rate_2_10,state	368 rows	v9 NICER (prefix)
data/v10/processed/nicer_full_features.csv	as v9 + bkg_ratio_12_15, split halves	368 rows	v10 NICER (full files; 366 pass QC)
data/v11/processed/nicer_new_features.csv	as v10 + lb*_se, group	318 rows	v11 never-used observations
data/v12/processed/bkg3c50.csv	B_*, f_A,f_B,f_C,f_T, status	503 rows	3C50 band rates and fractions
data/v12/processed/nicer_corrected_screened.csv	corrected c1,c2,T1..T3,STATE, f_used	534 rows	frozen v12 training set (used by v13)
data/v13/processed/nicer_new_features.csv; bkg3c50.csv	as v11; as v12	238; 148 rows	v13 third observation set

2.3.1 Worked example: the first rows of the real v2 matrices

Table 2.2 shows the first observations of data/v2/processed/features.npz (rows are stored source by source in the order of the source list, BH first) and the derived inputs of the representations. Table 2.3 shows the first rows of the NICER feature table. Figure 2.1 shows the whole 456×45 B matrix.

2.4 The design matrix and the leave-one-source-out split

Figure 2.2 shows schematically what one LOSO fold sees. The test block is *all* observations of one source; the training block is all observations of every other source. The standardisation parameters (μ, σ), imputation medians, sample weights and any selected hyper-parameter or threshold are computed from the training block only. After S folds every row has exactly one OOF score.

Table 2.2: First rows of the v2 inputs (read from `data/v2/processed/features.npz` and `data/v5/processed/timing_features.csv`). x_0, x_{44} : net rate in the first (4.908–5.315 keV) and last (24.151–25.008 keV) grid bins [c/s/keV/PCU]; $A_0 = \text{asinh}(x_0/0.1)$; $B_0 = x_0/F$; h_1, h_2 : colours (Eq. (1.11)); F in c/s/PCU. Note the slightly negative x_{44} of 91702-01-22-00 (a soft-state spectrum), kept as is.

obs_id	source_id	y	F	h_1	h_2	$\log_{10} F$	x_0	A_0	B_0	x_{44}	T_1	T_2
90428-01-01-07	GROJ1655-40	1	36.31	0.9495	0.3864	1.560	6.115	4.807	0.1684	0.2296	0.1124	0.1923
91702-01-02-03	GROJ1655-40	1	607.22	0.4126	0.2450	2.783	284.152	8.645	0.4680	0.7872	0.0172	0.0162
91702-01-22-00	GROJ1655-40	1	728.45	0.3094	0.0235	2.862	402.680	8.994	0.5528	−0.0098	0.0309	0.0237
91702-01-22-03	GROJ1655-40	1	793.68	0.3312	0.0310	2.900	421.505	9.040	0.5311	0.0034	0.0305	0.0259
... (rows 4–104: the other BH observations) ...												
50030-02-04-00	4U1636-53	0	183.57	0.6820	0.1084	2.264	55.527	7.013	0.3025	0.0765		
60032-01-20-000	4U1636-53	0	104.24	0.7208	0.1224	2.018	29.900	6.394	0.2868	0.0393		
91024-01-74-00	4U1636-53	0	80.50	0.8607	0.3250	1.906	16.855	5.820	0.2094	0.3327		

Table 2.3: First rows of `data/v10/processed/nicer_full_features.csv` (NICER, full files). $c_1 = C(4-6)/C(2-4)$, $c_2 = C(6-10)/C(4-6)$; T_k and “rms” are fractional rms (signed square roots, Eq. (1.20)); rms is the 0.1–10 Hz state index; rate in cs^{-1} (2–10 keV); n_{seg} = valid 128 s segments.

obsid	source	label	c_1	c_2	T_1	T_2	T_3	rms	ν_c [Hz]	rate	state	n_{seg}
0100320102	Cyg X-1	BH	0.1664	0.2826	0.0661	0.1351	0.1566	0.2276	1.124	2920.9	hard-like	12
2636010102	Cyg X-1	BH	0.0638	0.1857	0.0520	0.0645	0.0462	0.0843	0.378	3591.0	intermediate	10
4690020110	Cyg X-1	BH	0.2544	0.3400	0.1321	0.2211	0.1509	0.2837	0.560	757.9	hard-like	18
5100320107	Cyg X-1	BH	0.2517	0.3357	0.1113	0.2087	0.1589	0.2799	0.677	876.7	hard-like	20
5100320118	Cyg X-1	BH	0.2013	0.3062	0.0792	0.2133	0.1744	0.2990	1.057	1378.4	hard-like	16

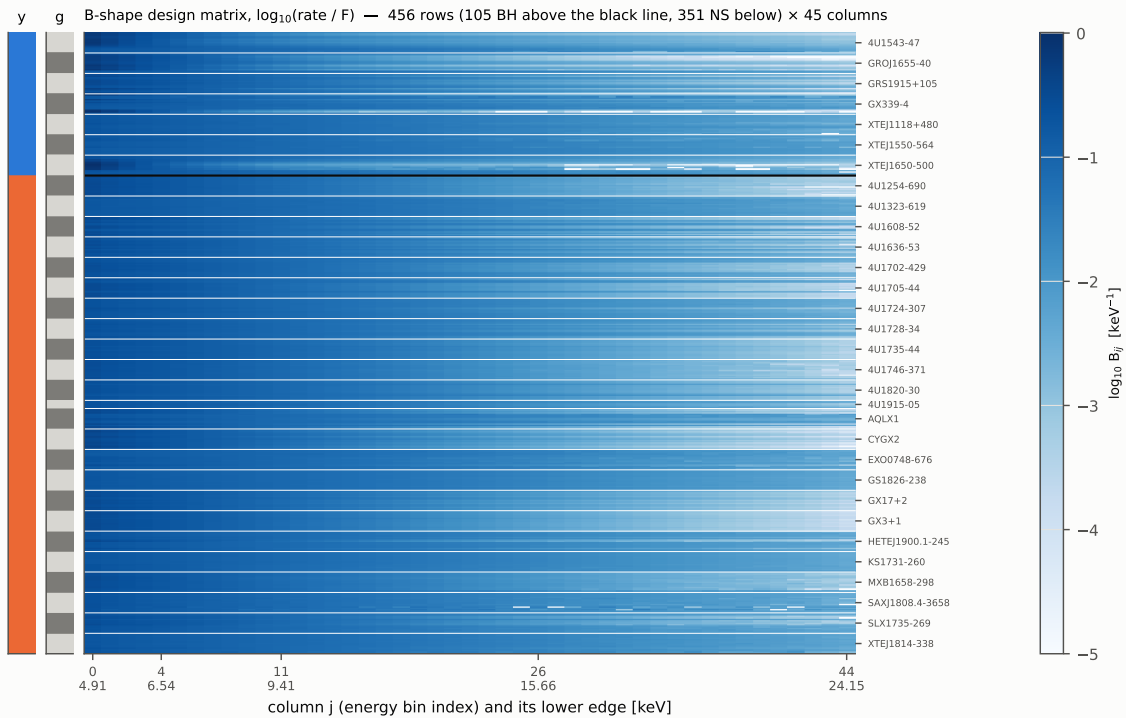


Figure 2.1: The real v2 design matrix for representation B: 456 rows (observations) $\times$ 45 columns (energy bins), colour = $\log_{10} B_{ij}$ (sequential scale, darker = larger). Rows are grouped by source (right labels; alternating grey in strip g), BH sources first (strip y : blue = 1) and NS below the black line (orange = 0). The steep fall of the shape with energy dominates every row; differences between sources are subtle and mostly in the high-energy columns, consistent with the finding that two colours capture the usable information. Generated for this report from `data/v2/processed/features.npz`.

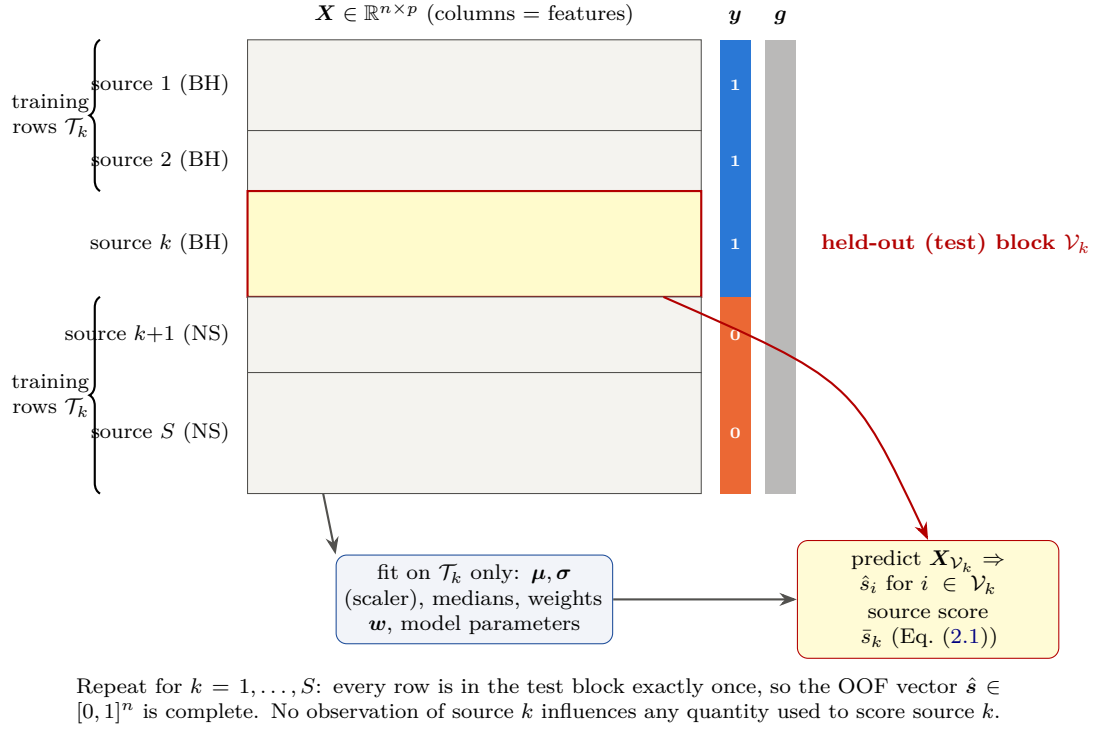


Figure 2.2: Schematic of the design matrix and one leave-one-source-out fold. Rows are observations grouped by source; the y column (blue = BH, orange = NS) and the source index g are attached to $\mathbf{X}$. The yellow block is the held-out source; every learned quantity – standardisation, imputation medians, sample weights, hyper-parameters, thresholds – is estimated from the grey training rows only. Drawn for this report.

2.5 End-to-end data flow

Figure 2.3 maps raw archives to reported numbers, with the script numbers of `scripts/`. Appendix ?? gives the complete script $\rightarrow$ output table.

2.6 Leakage controls implemented in the code

1. **Folds by source.** `LeaveOneGroupOut` on g ; every script asserts that each test fold contains exactly one source, that training and test sources are disjoint, and that both classes are present in training (`results/fold_checks.csv`, `results/v2/fold_checks.csv`).
2. **Fixed folds across versions.** The v2 folds (`results/v2/folds_loso.csv`) are re-used and asserted identical in v3a, v4a and v5, so paired comparisons use the same test sets.
3. **Train-only preprocessing.** `StandardScaler` is part of the model object (`make_pipeline` or `V4Model.fit`), imputation medians come from training rows (Eq. (2.2)), and source-equal weights are computed from training sources only.
4. **Nested selection.** Hyper-parameters, representations and thresholds that are chosen at all are chosen by an inner LOSO over the training sources (v4a, v7c2) – never on the test source (Section 4.4).
5. **Label-free residualisation.** The colour-conditional permutation tests regress timing on colour with a kNN that never sees labels, fitted leaving the source out (Section 4.6).

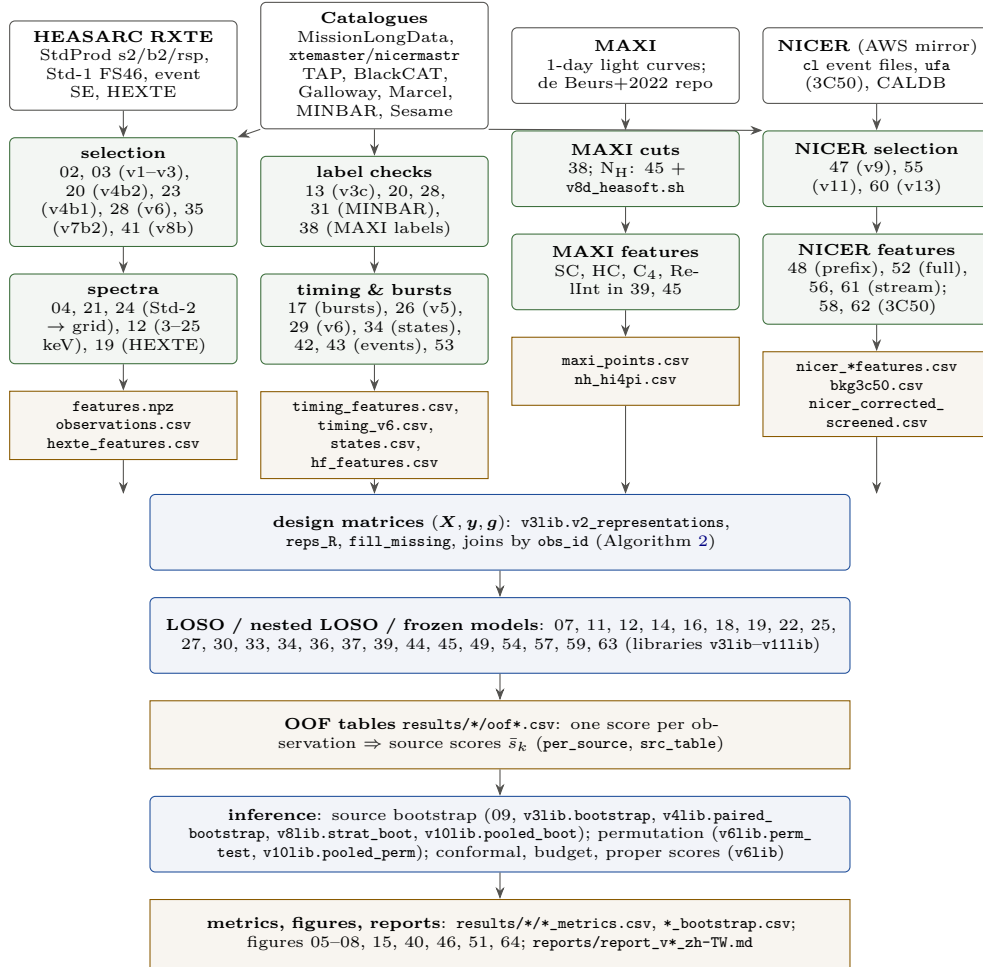


Figure 2.3: End-to-end data flow of the project. Grey boxes: external archives; green: selection and processing scripts (numbers = `scripts/NN_*.py`); amber: committed intermediate files; blue: model fitting and inference. Raw files (left of the green boxes) are not in git; everything from the amber row down is committed, so every number in this report can be traced to a file in `results/` or `data/`. Drawn for this report.

6. **Frozen models for external and prospective tests.** External (v6a, v8b) and new-observation (v11, v13) tests apply models whose parameters, imputation medians and decision rules were fixed before the test data were read; for the same-source prospective tests each source is scored by the model fitted *without* that source (“out-of-source”, Section 4.10).
7. **No position-based joins without assertion.** Feature tables are joined by `obs_id`; where arrays are concatenated by position (e.g. `observations.csv` and `features.npz`) the code asserts equality of the ID order.

Chapter 3

Machine-learning methods

Every model used in the project is defined here once, in the form in which it was actually run: scikit-learn 1.8.0 (`requirements-lock.txt`) with the settings of `scripts/config.py` and the wrappers `V4Model` (`v4lib.py`), `C2Model` (`v7lib.py`) and the pipelines of `07_train_eval.py` and `v3lib.py`. Equations describe the scikit-learn implementation, not a generic textbook variant. Table 3.1 is the inventory; `??` collects all hyper-parameters.

Table 3.1: Inventory of the models and where they are used. Role: P = model of a pre-registered primary comparison; B = baseline; D/E = descriptive/exploratory only.

Model	Versions	Code	Role
Dummy (prior)	v1, v2	<code>07_train_eval.py</code>	B
Logistic Regression	v1–v13	<code>07</code> , <code>v3lib.MODELS</code> , <code>V4Model</code> , <code>C2Model</code> , <code>CCTLR.lr</code>	P (v2–v11), B
Random Forest	v1–v13	as above	P (v8d, v12b, v13a), D
ExtraTrees	v4a	<code>V4Model</code>	E
HistGradientBoosting	v4a	<code>V4Model</code>	E
RBF SVM	v4a, v7c2; v7c1 (Platt, 3-class)	<code>V4Model</code> , <code>C2Model</code> , <code>37_v7c1_repro.py</code>	E / reproduction
kNN classifier	v4a, v7c2, v8d; v7c1 (3-class)	<code>V4Model</code> , <code>C2Model</code> , <code>37</code>	E / reproduction
QDA	v4a	<code>V4Model</code>	E
MLP (seed ensemble)	v4a	<code>V4Model</code>	E
Nested auto-selection	v4a 1d/1e	<code>v4lib.nested_fold</code> (Section 4.4)	P (v4)
CCTLR	v6c	<code>v6lib.CCTLR</code>	D (new method)
kNN regression	v6d, v9b–v13b	<code>v6lib.source_residuals</code>	inside the tests
Frozen ν^* rule	v11, v13	<code>57_v11_analysis.py</code> , <code>63_v13_analysis.py</code>	D

3.1 Shared components

3.1.1 Standardisation

For the scale-sensitive models (LR, SVM, kNN, QDA, MLP, the colour part of CCTLR and the kNN regression) each feature is standardised with the training rows $\mathcal{T}$ of the current fit only

(StandardScaler):

$$\mu_\ell = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} x_{i\ell}, \quad \sigma_\ell = \left(\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} (x_{i\ell} - \mu_\ell)^2 \right)^{1/2}, \quad z_{i\ell} = \frac{x_{i\ell} - \mu_\ell}{\sigma_\ell} \quad (\sigma_\ell = 0 \Rightarrow \text{scale } 1), \quad (3.1)$$

and the same (μ_ℓ, σ_ℓ) are applied to test rows. Tree models (RF, ExtraTrees, HistGB) use raw features. Standardisation is learned; the per-spectrum normalisations of Section 1.5 (B, H, ...) are not.

3.1.2 Class-balanced weights

The BH:NS ratio is 7:24 sources in the main sample. With n training rows of which n_c are in class c , `class_weight='balanced'` gives every row of class c the weight

$$w_i^{\text{bal}} = \frac{n}{2 n_{y_i}}, \quad \sum_{i: y_i=c} w_i^{\text{bal}} = \frac{n}{2}. \quad (3.2)$$

In scikit-learn these class weights multiply any sample weights inside the loss (LR, SVM, HistGB) or inside the impurity (trees). For QDA (no weights) the same balance is imposed through equal priors, for kNN through a prior correction of the output (Eq. (3.10)), and for the MLP through w^{bal} passed as sample weights.

Source-equal weights (v4a 1c, v4b1, v5 S3, v6 S3). When sources contribute very different numbers of observations (v4b1: 6 to 1329 per source), balanced weights let the most-observed sources dominate. With S_c the number of training sources in class c and n_k the training rows of source k (`v4lib.source_equal_weights`):

$$w_i^{\text{src}} = \frac{n}{2 S_{y_i} n_{g_i}}, \quad \sum_{i: g_i=k} w_i^{\text{src}} = \frac{n}{2 S_c} \quad (k \in \text{class } c), \quad \sum_{i: y_i=c} w_i^{\text{src}} = \frac{n}{2}. \quad (3.3)$$

Every source of a class has the same total weight, both classes sum to $n/2$ (the same overall scale as balanced weights), and $w^{\text{src}} = w^{\text{bal}}$ exactly when all sources have the same n_k (unit test in `test_v4lib.py`). When w^{src} is passed, `class_weight` is set to `None`.

Implementation fix: scale of the source-equal weights (2026-10-01, after the v4a results)

The first version normalised each class to total weight 1 (total 2 per fit) instead of $n/2$. Because the LR objective multiplies the summed weighted loss by C (Eq. (3.5)), this acted like an $\approx n/2$ times stronger L2 penalty (scores squeezed into 0.4–0.6); the HistGB internal `min_hessian_to_split` was also affected (RF is scale-invariant). The fix rescaled to $n/2$ per class; only v4a parts 1a–1c were re-run: 1a and 1b were bit-identical, 12 combinations of 1c changed, and no conclusion changed (pre-fix results in commit 9120afb). v4b1 was run only with the fixed weights.

3.1.3 Score conventions

Every model returns a BH score $s(\mathbf{x}) \in [0, 1]$: `predict_proba[:, 1]` for LR, RF, ExtraTrees, HistGB, QDA; the prior-corrected vote fraction for kNN; the seed-averaged probability for the MLP; the logistic function of the decision value for the SVM and CCTLR. Scores are rankings and thresholdable quantities, not calibrated probabilities, except where proper scoring rules are explicitly evaluated (v6g).

3.2 Dummy classifier

Role. Baseline in v1–v2 (07_train_eval.py). **Definition.** `DummyClassifier(strategy='prior')` returns the training class frequencies, $s(\mathbf{x}) = n_1/n$ for every $\mathbf{x}$, and predicts the majority class.

Why it matters. Under LOSO the majority class of the training fold depends on which source is held out: with 4 BH and 4 NS sources of 15 observations each (v1), leaving out a BH leaves 45 BH : 60 NS, so the held-out BH is predicted NS, and vice versa; the Dummy balanced accuracy is therefore 0, not 0.5. With 7 BH and 24 NS sources (v2) NS is always the majority and the Dummy balanced accuracy is 0.5. “Chance level” under LOSO is thus not a fixed number, one reason why AUC is also reported.

Algorithm 3 Dummy (prior) classifier

Input: training labels $\mathbf{y}_{\mathcal{T}}$; test rows $\mathcal{V}$

- 1: $\pi \leftarrow \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} y_i$ ▷ no features; only hyper-parameter `strategy='prior'`
 - 2: **return** $s_i = \pi$ and $\hat{y}_i = \mathbf{1}[\pi \geq 0.5]$ for every $i \in \mathcal{V}$
-

3.3 Logistic regression

Role. The workhorse: H-LR (two colours + LR) is the v2 reference against which v4 tests every alternative; H+T-LR is the primary model of v5, v6a, v8a/b, v9a, v10a, v11a. **Model.** On standardised inputs $\mathbf{z}$,

$$s(\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{z} + b), \quad \sigma(u) = \frac{1}{1 + e^{-u}}. \quad (3.4)$$

Objective (`LogisticRegression(C=1, penalty='l2', solver='lbfgs', max_iter=5000)`):

$$(\hat{\mathbf{w}}, \hat{b}) = \arg \min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i \in \mathcal{T}} \omega_i \left[-y_i \log s(\mathbf{x}_i) - (1 - y_i) \log(1 - s(\mathbf{x}_i)) \right], \quad (3.5)$$

with $\omega_i = w_i^{\text{bal}}$ (Eq. (3.2)) or w_i^{src} (Eq. (3.3)); the intercept is not penalised. scikit-learn minimises the equivalent $\frac{1}{\Omega} \sum_i \omega_i \ell_i + \frac{1}{2C\Omega} \|\mathbf{w}\|^2$ ($\Omega = \sum_i \omega_i$) with L-BFGS. $C = 1$ means the penalty weight 1/2 competes with a weighted log-loss summed over $\approx n$ rows: for $n \approx 400$ and 2–6 features the penalty is weak. **Decision boundary.** In raw colours the 0.5 boundary of H-LR is the line $\sum_\ell (w_\ell / \sigma_\ell) h_\ell + b - \sum_\ell w_\ell \mu_\ell / \sigma_\ell = 0$ (used to draw ??). **Assumptions.** Log-odds linear in the standardised features; a single monotone effect per feature. This is the main reason LR can use timing only as “more variability $\Rightarrow$ more BH” everywhere in the colour plane (motivating CCTLR, Section 3.11), and it fails on MAXI where BHs lie on both sides of the NS colour track (??).

Algorithm 4 LR fit and score (one fold)

Input: $\mathbf{X}_{\mathcal{T}}, \mathbf{y}_{\mathcal{T}}$, optional $\mathbf{w}$; test rows $\mathbf{X}_{\mathcal{V}}$

- 1: $(\boldsymbol{\mu}, \boldsymbol{\sigma}) \leftarrow$ Eq. (3.1) on $\mathbf{X}_{\mathcal{T}}$; $\mathbf{Z}_{\mathcal{T}}, \mathbf{Z}_{\mathcal{V}}$ standardised
 - 2: $\omega_i \leftarrow w_i^{\text{bal}}$ if $\mathbf{w}$ is None else w_i
 - 3: $(\mathbf{w}, b) \leftarrow$ L-BFGS minimisation of Eq. (3.5) (max. 5000 iterations)
 - 4: **return** $s_i = \sigma(\mathbf{w}^\top \mathbf{z}_i + b)$ for $i \in \mathcal{V}$
-

Table 3.2: Logistic-regression hyper-parameters (LR_PARAMS in config.py).

Parameter	Value	Fixed / tuned	Grid (where tuned)
C	1.0	fixed (all versions)	v4a 1d/1e: {0.01, 0.1, 1, 10} (V4_GRID['LogReg
penalty, solver	L2, lbfgs	fixed	–
class_weight	balanced (or None with w^{src})	fixed	–
max_iter	5000	fixed	–

3.4 Random forest

Role. Second reference model in every version; primary model of v8d (MAXI), v12b (chosen post hoc after v9–v11) and v13a (chosen before the v13 data). **Model.** An average of B classification trees,

$$s(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{p}_b(\mathbf{x}), \quad \hat{p}_b(\mathbf{x}) = \frac{\sum_{i \in \text{leaf}_b(\mathbf{x})} \omega_i^{(b)} y_i}{\sum_{i \in \text{leaf}_b(\mathbf{x})} \omega_i^{(b)}}, \quad (3.6)$$

where $\omega_i^{(b)} = w_i^{\text{bal}} \cdot m_i^{(b)}$ (or $w_i^{\text{src}} m_i^{(b)}$), and $m_i^{(b)}$ is the number of times row i is drawn in the bootstrap sample of tree b (n draws with replacement). The balanced class weights are computed once on the whole training fold, not per bootstrap sample. **Tree growth.** CART: at each node draw $m = \max(1, \lfloor \sqrt{p} \rfloor)$ candidate features without replacement and choose the split that maximises the weighted decrease of the Gini impurity

$$G(\mathcal{N}) = 1 - \sum_{c \in \{0,1\}} \pi_c(\mathcal{N})^2, \quad \pi_c(\mathcal{N}) = \frac{\sum_{i \in \mathcal{N}, y_i=c} \omega_i}{\sum_{i \in \mathcal{N}} \omega_i}, \quad \Delta = \Omega_{\mathcal{N}} G(\mathcal{N}) - \Omega_L G(L) - \Omega_R G(R); \quad (3.7)$$

grow without depth limit until nodes are pure or a child would have fewer than `min_samples_leaf` rows. With the two colours of H, $m = 1$: every split uses one randomly chosen colour; with H plus three timing features and a missing indicator ($p = 6$), $m = 2$. **Determinism. random_state=42.** **Assumptions/limitations.** Piecewise-constant, can follow non-monotone class regions (important for MAXI and for the timing×colour interaction), but with < 10 BH sources a leaf can memorise one source; scores of trees are not calibrated (v3: A-RF had good ranking but scores pushed below 0.5 – a threshold problem).

Algorithm 5 Random forest fit and score (one fold)

Input: $\mathbf{X}_{\mathcal{T}}, \mathbf{y}_{\mathcal{T}}$ (raw features), ω_i (balanced or source-equal), B , m , leaf size L

- 1: **for** $b = 1, \dots, B$ **do** ▷ seeded
 - 2: draw bootstrap counts $m^{(b)} \sim \text{Multinomial}(n; 1/n, \dots, 1/n)$; weights $\omega_i^{(b)} = \omega_i m_i^{(b)}$
 - 3: grow tree b : recursively split nodes on the best of m random features by Eq. (3.7), leaves $\geq L$ rows
 - 4: **end for**
 - 5: **return** s_i by Eq. (3.6) for $i \in \mathcal{V}$
-

3.5 Extremely randomised trees

Role. Exploratory alternative in v4a. **Difference from RF.** No bootstrap ($m_i^{(b)} = 1$), and at each node, for each of the m candidate features one threshold is drawn uniformly between the feature’s minimum and maximum in the node; the best of these random splits by Eq. (3.7) is used. Settings: 500 trees, `max_features='sqrt'`, `min_samples_leaf=2`, `balanced`, `random_`

Table 3.3: Random-forest hyper-parameters (RF_PARAMS, V4_INNER_TREES, V4_OUTER_TREES).

Parameter	Value	Fixed / tuned	Grid / note
n_estimators	500	fixed	v4a nested: 200 trees in the inner loop, 500 for
max_features	'sqrt'	fixed	$m = \lfloor \sqrt{p} \rfloor$
min_samples_leaf	2	fixed	v4a 1d/1e: {1, 2, 5, 10}
class_weight	balanced (or None with w^{src})	fixed	–
bootstrap, criterion	True, Gini	default	–
random_state	42	fixed	n_jobs: 2 (v1–v3), 1 (V4Model)

state=42; nested grid {1, 2, 5, 10} for the leaf size (200/500 trees inner/outer). Lower variance than RF at the price of more bias; no scaling needed.

Algorithm 6 ExtraTrees fit and score

Input: raw $X_{\mathcal{T}}$, $y_{\mathcal{T}}$, weights ω_i ; $B = 500$ (inner 200), $m = \lfloor \sqrt{p} \rfloor$, leaf size L

- 1: **for** $b = 1, \dots, B$ **do** ▷ all rows, no bootstrap
 - 2: at each node: draw m features, one threshold $\sim U(\text{min}, \text{max})$ each; keep the split with the largest Δ (Eq. (3.7)); stop at purity or leaf size L
 - 3: **end for**
 - 4: **return** $s_i = \frac{1}{B} \sum_b \hat{p}_b(\mathbf{x}_i)$ (Eq. (3.6) with $m_i^{(b)} = 1$)
-

Table 3.4: ExtraTrees hyper-parameters (V4_FIXED['ExtraTrees'], V4_GRID['ExtraTrees']).

Parameter	Value	Fixed / tuned	Grid
n_estimators	500	fixed (inner 200)	–
max_features	'sqrt'	fixed	–
min_samples_leaf	2	tuned in 1d/1e	{1, 2, 5, 10}
class_weight, random_state	balanced, 42	fixed	–

3.6 Histogram gradient boosting

Role. Exploratory in v4a. **Model.** An additive model of M regression trees on the log-odds scale, $F_M(\mathbf{x}) = F_0 + \eta \sum_{m=1}^M f_m(\mathbf{x})$, $s(\mathbf{x}) = \sigma(F_M(\mathbf{x}))$, with F_0 the weighted log-odds of the training classes. Each tree f_m is fitted to the weighted gradients and Hessians of the binary log loss at F_{m-1} , $g_i = \omega_i (\sigma(F_{m-1}(\mathbf{x}_i)) - y_i)$, $h_i = \omega_i \sigma(1 - \sigma)$; leaf values are $-\sum_{i \in \text{leaf}} g_i / (\sum_{i \in \text{leaf}} h_i + \lambda)$ with $\lambda = 12_regularization = 0$; trees are grown best-first on features pre-binned into at most 255 quantile bins, up to max_leaf_nodes leaves with at least min_samples_leaf rows. Settings (V4_FIXED['HistGB']): $\eta = 0.1$, $M = 200$, 15 leaves, 20 rows per leaf, no early stopping, balanced, random_state=42; nested grid $\eta \in \{0.05, 0.1\} \times \text{leaves} \in \{7, 15\}$. **Limitation.** With 30 training sources and 20 rows per leaf, a single source (15 rows) cannot form a leaf, which partly protects against memorisation; with 2 colours the model is close to a smooth version of RF.

3.7 Support-vector machine with RBF kernel

Role. Exploratory in v4a (it was selected in 23 of 31 outer folds of the nested auto-selection) and descriptive in v7c2; a Platt-scaled three-class version reproduces ? in v7c1. **Training**

Algorithm 7 HistGradientBoosting fit and score**Input:** raw $\mathbf{X}_{\mathcal{T}}$, $\mathbf{y}_{\mathcal{T}}$, weights ω_i ; η , M , max leaves, min leaf rows

- 1: bin every feature into ≤ 255 quantile bins (training rows); $F_0 \leftarrow \text{logit}(\sum \omega_i y_i / \sum \omega_i)$
- 2: **for** $m = 1, \dots, M$ **do**
- 3: $g_i \leftarrow \omega_i(\sigma(F_{m-1}(\mathbf{x}_i)) - y_i)$, $h_i \leftarrow \omega_i \sigma(1 - \sigma)$
- 4: grow a best-first tree on (g, h) ; leaf value $-\sum g / (\sum h + \lambda)$; $F_m \leftarrow F_{m-1} + \eta f_m$
- 5: **end for**
- 6: **return** $s_i = \sigma(F_M(\mathbf{x}_i))$

Table 3.5: HistGradientBoosting hyper-parameters (V4_FIXED['HistGB'], V4_GRID['HistGB']).

Parameter	Value	Fixed / tuned	Grid
learning_rate η	0.1	tuned in 1d/1e	{0.05, 0.1}
max_leaf_nodes	15	tuned in 1d/1e	{7, 15}
max_iter M	200	fixed	—
min_samples_leaf	20	fixed	—
l2_regularization λ	0	fixed	—
early_stopping	False	fixed	—
class_weight, random_state	balanced, 42	fixed	—

(libsvm dual, labels $y'_i = 2y_i - 1$, standardised inputs $\mathbf{z}$):

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y'_i y'_j K(\mathbf{z}_i, \mathbf{z}_j) \quad \text{s.t. } 0 \leq \alpha_i \leq C c_{y_i}, \sum_i \alpha_i y'_i = 0, \quad K(\mathbf{z}, \mathbf{z}') = e^{-\gamma \|\mathbf{z} - \mathbf{z}'\|^2}, \quad (3.8)$$

with class weights $c_y = n/(2n_y)$. **Kernel width.** `gamma='scale'` sets $\gamma = 1/(p \text{Var}(\mathbf{Z}))$, where $\text{Var}(\mathbf{Z})$ is the variance of *all* entries of the standardised training matrix (≈ 1 , so $\gamma \approx 1/p$); in the v4a grid $\gamma = \kappa/(p \text{Var}(\mathbf{Z}))$ with factor $\kappa \in \{0.3, 1, 3\}$ (code `gamma_factor`). **Score.** The decision value $f(\mathbf{z}) = \sum_i \alpha_i y'_i K(\mathbf{z}_i, \mathbf{z}) + b$ is mapped by the logistic function,

$$s(\mathbf{x}) = \sigma(f(\mathbf{z})), \quad (3.9)$$

so $s = 0.5$ is exactly the SVM decision surface; Platt scaling is deliberately not used because it runs an internal random cross-validation. **v7c2 variant.** $C \in \{0.1, 1, 10\} \times \gamma \in \{0.1, 0.585, 3\}$ (absolute γ), standardised, balanced, same score; selected by inner grouped 5-fold source AUC (Section 4.4). **v7c1 variant.** Three classes (BH, non-pulsing NS, pulsar), one-vs-one, $C = 0.655298$, $\gamma = 0.585144$ on training-standardised features, unweighted, Platt probabilities (`probability=True`, `random_state=42`), exactly as the R/e1071 setting of ?.

Algorithm 8 RBF-SVM fit and score (binary versions)**Input:** $\mathbf{X}_{\mathcal{T}}$, $\mathbf{y}_{\mathcal{T}}$; C ; γ or factor κ

- 1: standardise (Eq. (3.1)); if 'scale': $\gamma \leftarrow \kappa/(p \text{Var}(\mathbf{Z}_{\mathcal{T}}))$ ($\kappa = 1$ for the fixed model)
- 2: solve the dual Eq. (3.8) with box constraints $C c_{y_i}$ (libsvm SMO)
- 3: **return** $s_i = \sigma(f(\mathbf{z}_i))$ (Eq. (3.9))

3.8 k -nearest-neighbour classifier

Role. Exploratory (v4a), descriptive (v7c2, v8d), reproduction (v7c1). **Model.** With $\mathcal{N}_k(\mathbf{z})$ the k training rows nearest to $\mathbf{z}$ in Euclidean distance on standardised features (uniform weights),

Table 3.6: SVM hyper-parameters (`V4_FIXED['SVM_RBF']`, `V4_GRID`, `V7_C2_SVM_GRID`, `V7_DEBEURS_SVM`).

Use	C	γ	selection
v4a fixed (1b)	1	'scale'	–
v4a nested (1d/1e)	{0.1, 1, 10}	{0.3, 1, 3} × 'scale'	inner LOSO, source AUC
v7c2 (MAXI)	{0.1, 1, 10}	{0.1, 0.585, 3}	inner grouped 5-fold, source AUC
v7c1 (reproduction)	0.655298	0.585144	fixed (from the paper), Platt, 3 classes

$\hat{p}(\mathbf{x}) = \frac{1}{k} \sum_{i \in \mathcal{N}_k} y_i$. Because the training folds are imbalanced, the output is corrected to equal class priors with $\pi = n_1/n$ of the training fold (`V4Model.score`, `C2Model.score`):

$$s(\mathbf{x}) = \frac{\hat{p}/\pi}{\hat{p}/\pi + (1 - \hat{p})/(1 - \pi)}. \quad (3.10)$$

Hyper-parameters. $k = 15$ fixed (v4a 1b); grids {5, 15, 31} (v4a nested) and {5, 15, 24, 35} (v7c2, v8d, inner LOSO); k is capped at the number of training rows. **v7c1.** Three classes, $k = 24$, *raw* (unstandardised) features and plain vote fractions, as ?. **Limitation.** A test source’s nearest neighbours can all come from one training source of similar brightness or colour; this is exactly how the model can “recognise” a source type rather than a compact-object type.

Algorithm 9 kNN classifier (binary, prior-corrected)

Input: $\mathbf{X}_{\mathcal{T}}$, $\mathbf{y}_{\mathcal{T}}$, k

- 1: standardise (Eq. (3.1)); $\pi \leftarrow$ fraction of BH rows in $\mathcal{T}$; $k \leftarrow \min(k, |\mathcal{T}|)$
 - 2: **for** $i \in \mathcal{V}$ **do**
 - 3: $\hat{p} \leftarrow$ BH fraction among the k nearest training rows; $s_i \leftarrow$ Eq. (3.10)
 - 4: **end for**
-

Table 3.7: kNN hyper-parameters (`V4_FIXED['kNN']`, `V4_GRID['kNN']`, `V7_C2_KNN_GRID`, `V7_DEBEURS_KNN_K`).

Use	k	scaling	output	selection
v4a fixed	15	standardised	prior-corrected	–
v4a nested	{5, 15, 31}	standardised	prior-corrected	inner LOSO, source AUC
v7c2, v8d	{5, 15, 24, 35}	standardised	prior-corrected	inner LOSO, source AUC
v7c1	24	raw	3-class vote fractions	fixed (paper)

3.9 Quadratic discriminant analysis

Role. Exploratory (v4a). **Model.** Class-conditional Gaussians on standardised features with regularised covariances and equal priors $\pi_0 = \pi_1 = \frac{1}{2}$:

$$\hat{\Sigma}_c = (1 - r) S_c + r \mathbf{I}, \quad s(\mathbf{x}) = \frac{\pi_1 \mathcal{N}(\mathbf{z}; \hat{\boldsymbol{\mu}}_1, \hat{\Sigma}_1)}{\sum_c \pi_c \mathcal{N}(\mathbf{z}; \hat{\boldsymbol{\mu}}_c, \hat{\Sigma}_c)}, \quad (3.11)$$

where S_c is the unbiased sample covariance of class c (scikit-learn applies the shrinkage to the squared singular values of the centred class data, which is equivalent). $r = 0.1$ fixed; nested grid {0.01, 0.1, 0.5} (`V4_GRID['QDA']`). QDA takes no sample weights, so class balance enters only through the priors. **Note.** Three QDA combinations were the only v4a combinations whose source balanced-accuracy interval excluded 0 (??); with 76 combinations this is within the expected number of chance exceedances.

Algorithm 10 QDA fit and score**Input:** $\mathbf{X}_{\mathcal{T}}, \mathbf{y}_{\mathcal{T}}, r$

- 1: standardise; for $c \in \{0, 1\}$: class mean $\hat{\boldsymbol{\mu}}_c$, unbiased covariance S_c , $\hat{\Sigma}_c \leftarrow (1 - r)S_c + r\mathbf{I}$
- 2: **return** s_i by Eq. (3.11) with $\pi_0 = \pi_1 = \frac{1}{2}$

Table 3.8: QDA hyper-parameters (`V4_FIXED['QDA']`, `V4_GRID['QDA']`).

Parameter	Value	Fixed / tuned	Grid
<code>reg_param</code> r	0.1	tuned in 1d/1e	$\{0.01, 0.1, 0.5\}$
<code>priors</code>	$[0.5, 0.5]$	fixed	—

3.10 Multi-layer perceptron (seed ensemble)

Role. Exploratory (v4a). **Model.** One hidden layer of 32 ReLU units (grid: (16) or (32, 16)), logistic output, on standardised inputs. **Objective** (per minibatch $\mathcal{M}$; scikit-learn 1.8 with sample weights):

$$\mathcal{L} = \frac{\sum_{i \in \mathcal{M}} \omega_i \ell_i}{\sum_{i \in \mathcal{M}} \omega_i} + \frac{\alpha}{2} \frac{\sum_W \|W\|_F^2}{\sum_{i \in \mathcal{M}} \omega_i}, \quad \omega_i = w_i^{\text{bal}}, \quad (3.12)$$

with ℓ_i the binary cross-entropy, $\alpha = 10^{-3}$ (grid $\{10^{-4}, 10^{-2}\}$); Adam (learning rate 10^{-3}), batch size $\min(200, n)$, at most 500 epochs. **Early stopping.** A stratified 20% validation split (seeded with the model seed) is held out; training stops when the weighted validation accuracy has not improved by 10^{-4} for 10 epochs, and the weights with the best validation accuracy are restored. **Seed ensemble.** Because a small MLP is unstable, the score is the mean of `predict_proba` over seeds $\{0, \dots, 4\}$ (outer fits) or $\{0, 1, 2\}$ (inner fits of the nested procedure): $s(\mathbf{x}) = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} s_r(\mathbf{x})$. A planned 1D-CNN on representation B was skipped as pre-declared because `torch` was not installed.

Algorithm 11 MLP seed ensemble**Input:** $\mathbf{X}_{\mathcal{T}}, \mathbf{y}_{\mathcal{T}}$; architecture, α ; seed set $\mathcal{R}$ (outer $\{0.4\}$, inner $\{0, 1, 2\}$)

- 1: standardise; $\omega_i \leftarrow w_i^{\text{bal}}$ (Eq. (3.2))
- 2: **for** $r \in \mathcal{R}$ **do**
- 3: stratified 80/20 split (seed r); Adam on Eq. (3.12); after each epoch the weighted validation accuracy; stop after 10 epochs without improvement $> 10^{-4}$ (max 500); restore the best weights
- 4: **end for**
- 5: **return** $s_i = \frac{1}{|\mathcal{R}|} \sum_r s_r(\mathbf{x}_i)$

3.11 Colour-conditional timing likelihood-ratio classifier (CCTLR, v6c)

Motivation. In LR, a timing feature has one slope everywhere in the colour plane, although the timing information is expected only where the colours overlap (hard states). CCTLR adds to the colour logit a generative log-likelihood ratio of the timing features *conditional on colour*, so that timing contributes only where the class-conditional timing distributions differ, and missing timing is simply “no evidence”. It was designed for v6 and has no tunable hyper-parameter.

Table 3.9: MLP hyper-parameters (`V4_FIXED['MLP']`, `V4_GRID['MLP']`, `V4_MLP_SEEDS_*`).

Parameter	Value	Fixed / tuned	Grid
<code>hidden_layer_sizes</code>	(32,)	tuned in 1d/1e	{(16,), (32, 16)}
<code>alpha</code>	10^{-3}	tuned in 1d/1e	$\{10^{-4}, 10^{-2}\}$
<code>early_stopping, validation_fraction</code>	True, 0.2	fixed	–
<code>max_iter</code>	500	fixed	–
activation, solver, learning rate, batch	ReLU, Adam, 10^{-3} , min(200, n)	defaults	–
seeds	outer $\{0, \dots, 4\}$, inner $\{0, 1, 2\}$	fixed	–

Model (`v6lib.CCTLR`): with standardised colours $\mathbf{h}$ and timing vector $\mathbf{T} = (T_1, T_2, T_3)$,

$$\ell(\mathbf{x}) = \underbrace{\mathbf{w}^\top \mathbf{h} + b}_{\text{H-LR logit}} + \Lambda(\mathbf{T} \mid \mathbf{h}), \quad s(\mathbf{x}) = \sigma(\ell(\mathbf{x})), \quad (3.13)$$

$$\Lambda(\mathbf{T} \mid \mathbf{h}) = \sum_{k=1}^3 \left[\log t_4(T_k; \mu_{1k}(\mathbf{h}), \tau_{1k}) - \log t_4(T_k; \mu_{0k}(\mathbf{h}), \tau_{0k}) \right], \quad \mu_{ck}(\mathbf{h}) = \beta_{ck0} + \beta_{ck1}h_1 + \beta_{ck2}h_2,$$

where $t_4(\cdot; \mu, \tau)$ is the Student- t density with 4 degrees of freedom, location μ and scale τ (robust to burst or dip outliers). For each class c and timing feature k , β_{ck} is a weighted least-squares fit on the training rows of class c with finite timing, with source-equal weights within the class $v_i = 1/(S_c n_{g_i})$ (`source_equal_within_class`) and a ridge of 10^{-9} ; the scale is the weighted residual standard deviation, floored at 0.01:

$$\hat{\beta}_{ck} = (\mathbf{Z}_c^\top \mathbf{V}_c \mathbf{Z}_c + 10^{-9} \mathbf{I})^{-1} \mathbf{Z}_c^\top \mathbf{V}_c \mathbf{T}_{c,k}, \quad \hat{\tau}_{ck} = \max\left(0.01, \sqrt{\sum_i v_i r_{ik}^2 / \sum_i v_i}\right), \quad (3.14)$$

with $\mathbf{Z}_c = [\mathbf{1}, \mathbf{h}]$ of class c and residuals r_{ik} . The LR part uses `LR_PARAMS` (balanced, or the source-equal weights of Eq. (3.3) in S3). If any timing value of a row is missing, $\Lambda = 0$ and the score equals the H-LR probability (unit test in `test_v6lib.py`). **Limitation.** Adding a generative LR term to a discriminative logit double counts nothing only if timing is conditionally independent of the label given colour under the LR model – an idealisation; the colour model of the class means is linear. In practice CCTLR did not beat H+T-LR (??).

Algorithm 12 CCTLR fit and score

Input: training colours $\mathbf{H}$, timing $\mathbf{T}$ (may contain NaN), labels $\mathbf{y}$, sources $\mathbf{g}$; test $(\mathbf{H}', \mathbf{T}')$

- 1: standardise $\mathbf{H}$ (Eq. (3.1)); fit H-LR (Eq. (3.5)) on $\mathbf{h}$
 - 2: **for** $c \in \{0, 1\}$ and $k \in \{1, 2, 3\}$ **do**
 - 3: rows: class c with all timing finite; weights $v_i = 1/(S_c n_{g_i})$; fit $\hat{\beta}_{ck}, \hat{\tau}_{ck}$ by Eq. (3.14)
 - 4: **end for**
 - 5: **for** each test row **do**
 - 6: $\Lambda \leftarrow 0$ if any T'_k missing, else the sum in Eq. (3.13)
 - 7: $s \leftarrow \sigma(\mathbf{w}^\top \mathbf{h}' + b + \Lambda)$
 - 8: **end for**
-

3.12 k -nearest-neighbour regression for colour residualisation

Role. Inside every colour-conditional permutation test (`v6d`, `v9b`, `v10b`, `v11b`, `v12a`, `v13b`; Section 4.6). It removes the part of a timing feature that is predictable from colour *without using labels*. For a feature F (e.g. $\log_{10} \nu_c$) and colours $\mathbf{h}$ standardised on the fit set (`v6lib.source_`

`residuals`):

$$\hat{F}_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(\mathbf{h}_i; \{j: g_j \neq g_i\})} F_j, \quad r_i = F_i - \hat{F}_i, \quad \bar{r}_k = \frac{1}{|\{i: g_i = k\}|} \sum_{i: g_i = k} r_i, \quad (3.15)$$

with $K = 25$ (`V6_KNN`; capped at the number of available rows) and uniform weights; the neighbours of an observation are taken only from *other* sources (leave-one-source-out), so a source cannot explain itself. Rows with missing F are skipped. For the v6 external replication the regression is fitted on all S1 rows and applied to the external rows. **Caveat** (noted in v9): when BHs cluster in one colour region, their neighbours are partly other BHs, so part of the class difference is absorbed into $\hat{F}$; the test is then conservative.

Algorithm 13 Label-free colour residualisation (`source_residuals`)

Input: colours $\mathbf{H}$, feature F (may be NaN), sources $\mathbf{g}$, $K = 25$

- 1: standardise $\mathbf{H}$ on the fit set
 - 2: **for** each source k **do**
 - 3: fit a kNN regression (K neighbours) on rows with finite F and $g \neq k$; predict rows of k ;
 $r_i \leftarrow F_i - \hat{F}_i$
 - 4: **end for**
 - 5: **return** source means $\bar{r}_k$ (Eq. (3.15))
-

3.13 Frozen characteristic-frequency rule (v11, v13)

A deliberately simple, pre-frozen decision rule: an observation is called BH if it is hard-like and $\nu_c < \nu^*$; a source is called BH if at least half of its new hard-like observations are called BH. The threshold is the geometric mean of the class medians of ν_c in the previous data set,

$$\nu^* = \sqrt{\text{median}_{\text{BH}} \nu_c \cdot \text{median}_{\text{NS}} \nu_c}, \quad (3.16)$$

frozen before any new data were read: $\nu^* = \sqrt{0.5917 \times 2.8594} = 1.3007$ Hz from v10 (32 BH and 111 NS hard-like observations; `logs/v11/nu_star_frozen.txt`) and $\nu^* = \sqrt{0.6294 \times 2.8362} = 1.3361$ Hz from the v12 corrected and screened data (62 BH, 138 NS; `logs/v13/nu_star_frozen.txt`).

3.14 Models that were not used

Principal component analysis was never used: every occurrence of “PCA” in the code refers to the RXTE Proportional Counter Array. The pre-declared 1D convolutional network on representation B (v4) was skipped because `torch` was not available; no replacement was introduced. No probability calibration (Platt, isotonic) was used except in the de Beurs reproduction (v7c1).

Chapter 4

Evaluation and statistical inference

This chapter defines how every number in the result tables is produced: the cross-validation scheme, the source-level decision, the metrics, the source bootstrap and its paired differences, the permutation tests, conformal prediction, the observation-budget curves and the protocol of the frozen (prospective) tests. Table 4.1 is a summary.

Table 4.1: Summary of the inferential procedures. “Unit” is what is resampled or permuted; all procedures keep the out-of-fold scores or the features fixed (no refitting inside the resampling).

Procedure		Statistic	Unit, size	Decision rule / code
Class-stratified bootstrap	source	BA, AUC (source, observation), Brier, log loss	sources within class; 2000, seed 42	95% percentile CI; paired difference “detected” iff its CI excludes 0 (<code>v4lib.paired_bootstrap</code>)
State-stratified bootstrap	boot-strap	observation within a state	AUC sources of the whole sample; 2000	as above (<code>v8lib.strat_boot</code>)
Pooled bootstrap (v10a)		equal-weight mean of component AUC gains	union of sources; 2000	as above (<code>v10lib.pooled_boot</code>)
Colour-conditional permutation	per- mutation	$D = \text{BH} - \text{NS}$ mean residual	source labels; 20 000, seed 42	two-sided $p < 0.05$; Holm across features (<code>v6lib.perm_test</code> , <code>holm</code>)
Pooled permutation (v10b)	permutation	mean of instrument D	labels within membership strata; 20 000	$p < 0.05$ (<code>v10lib.pooled_perm</code>)
Mondrian (v6e)	conformal	per-class p -values	calibration sources	set = $\{y : p_y > \varepsilon\}$, $\varepsilon \in \{0.1, 0.2\}$ (<code>v6lib.conformal_*</code>)
Observation (v6f)	budget	source BA/AUC with k obs/source	observations within source; 500 draws	median and 2.5–97.5% (<code>v6lib.budget_curves</code>)

4.1 Leave-one-source-out cross-validation

Folds are produced by `LeaveOneGroupOut` on $\mathbf{g}$; fold k (in the sorted order of the source names) holds out all observations of source k . The OOF score of observation i is

$$\hat{s}_i = s(\mathbf{x}_i; \hat{\theta}^{(-g_i)}), \quad \hat{\theta}^{(-k)} = \text{Fit}(\{(\mathbf{x}_j, y_j, g_j) : g_j \neq k\}), \quad (4.1)$$

where `Fit` includes every learned step of Algorithm 2. Each script asserts per fold: exactly one test source, disjoint training and test sources, and both classes in training (Section 2.6).

Algorithm 14 Leave-one-source-out OOF scores (`v3lib.run_loso`, `v4lib.loso_scores`)**Input:** $(\mathbf{X}, \mathbf{y}, \mathbf{g})$, model constructor $\mathcal{M}$, weighting option

- 1: **for** each source k (sorted) **do**
- 2: $\mathcal{V} \leftarrow \{i : g_i = k\}$, $\mathcal{T} \leftarrow \{i : g_i \neq k\}$; assert $|\{g_i : i \in \mathcal{V}\}| = 1$, both classes in $\mathcal{T}$
- 3: fill missing values with $\mathcal{T}$ -medians + indicator (Eq. (2.2)) if needed
- 4: $\mathbf{w} \leftarrow$ balanced or source-equal weights from $(\mathbf{y}_{\mathcal{T}}, \mathbf{g}_{\mathcal{T}})$
- 5: $m \leftarrow \mathcal{M}.\text{fit}(\mathbf{X}_{\mathcal{T}}, \mathbf{y}_{\mathcal{T}}, \mathbf{w})$; $\hat{s}_i \leftarrow m.\text{score}(\mathbf{x}_i)$ for $i \in \mathcal{V}$
- 6: **end for**
- 7: **return** $\hat{\mathbf{s}}$ and the source scores $\bar{s}_k$ (Eq. (2.1))

Leaky observation-wise split (diagnostic only). To show the size of the leak, `v1` and `v2` also ran a stratified 5-fold split over observations (`StratifiedKfold(5, shuffle=True, random_state=42)`), in which a source appears in both training and test folds. It is never used for any conclusion other than the size of the optimism (??).

4.2 Source-level aggregation and thresholds

The source score $\bar{s}_k$ is the mean of the OOF scores of its observations (Eq. (2.1)); the decision uses $t = 0.5$. In the nested procedures of `v4a` a threshold is chosen inside the training sources by maximising the inner source balanced accuracy (`v4lib.choose_threshold`). With $u_1 < \dots < u_q$ the distinct inner source scores and candidates $\mathcal{C} = \{u_1 - 10^{-9}\} \cup \{(u_r + u_{r+1})/2\}_{r=1}^{q-1} \cup \{u_q + 10^{-9}\}$:

$$t^* = \arg \min_{t \in \mathcal{C}^*} |t - 0.5|, \quad \mathcal{C}^* = \left\{ t \in \mathcal{C} : \text{BA}_{\text{src}}(t) = \max_{t' \in \mathcal{C}} \text{BA}_{\text{src}}(t') \right\}, \quad (4.2)$$

i.e. ties are broken towards 0.5 (unit tests: separable scores (0.1, 0.2, 0.6, 0.7) give $t = 0.4$). The outer scores are stored as $\hat{s}_i = \hat{s}_i^{\text{raw}} - t_k^* + 0.5$ so that the common rule $\hat{s} \geq 0.5$ applies; the outer *source AUC* of a nested procedure is computed on these shifted scores, an approximation because different folds may use different models (stated in the `v4` pre-registration).

4.3 Metrics

Balanced accuracy. With BH recall (sensitivity) and NS recall (specificity),

$$\text{BA} = \frac{1}{2}(\text{TPR} + \text{TNR}). \quad (4.3)$$

Source level: computed on $(\hat{Y}_k, Y_k)$ over sources; with 7 BH and 24 NS sources, “ k of 31 correct” is misleading (all-NS gives 24/31) and BA is the headline. *Observation level:* over observations; inside the bootstrap it is computed from per-source counts c_k (correct observations) and n_k as $\text{BA}_{\text{obs}} = \frac{1}{2}(\sum_{k \in \text{BH}} c_k / \sum_{k \in \text{BH}} n_k + \sum_{k \in \text{NS}} c_k / \sum_{k \in \text{NS}} n_k)$, which is the pooled recall of the resampled multiset of sources.

ROC AUC (Mann–Whitney form, ties counted one half). For positive scores $\{a_r\}_{r=1}^{n_1}$ and negative scores $\{b_q\}_{q=1}^{n_0}$,

$$\text{AUC} = \frac{1}{n_1 n_0} \sum_{r=1}^{n_1} \sum_{q=1}^{n_0} \left[\mathbf{1}[a_r > b_q] + \frac{1}{2} \mathbf{1}[a_r = b_q] \right]. \quad (4.4)$$

Source AUC uses $a = \{\bar{s}_k\}_{k \in \text{BH}}$, $b = \{\bar{s}_k\}_{k \in \text{NS}}$: the probability that a random BH source scores above a random NS source; it does not depend on the threshold. *Observation AUC* (used

inside states) uses all observations. With observation weights w (`v7lib.auc_ba`), $AUC_w = \sum_{r,q} w_r w_q [\mathbf{1}[a_r > b_q] + \frac{1}{2} \mathbf{1}[a_r = b_q]] / (\sum_r w_r \sum_q w_q)$; the reported state-stratified AUCs are unweighted.

Proper scores (v6g). Source-level, class-balanced, with the source score clipped to $[0.01, 0.99]$ (`v6lib.proper_score_bootstrap`):

$$LL = \frac{1}{2} \left[-\log \overline{\tilde{s}_k}_{k \in BH} + -\log(1 - \overline{\tilde{s}_k}_{k \in NS}) \right], \quad \text{Brier} = \frac{1}{2} \left[\overline{(1 - \tilde{s}_k)^2}_{k \in BH} + \overline{\tilde{s}_k^2}_{k \in NS} \right], \quad (4.5)$$

lower is better. Unlike AUC they cannot saturate at a perfect ranking, which matters in v6a.

4.4 Nested leave-one-source-out selection

When a hyper-parameter, a representation, an algorithm or a threshold is chosen, the choice is made inside each outer training set by an inner LOSO over its sources (Figure 4.1). The outer test source never takes part.

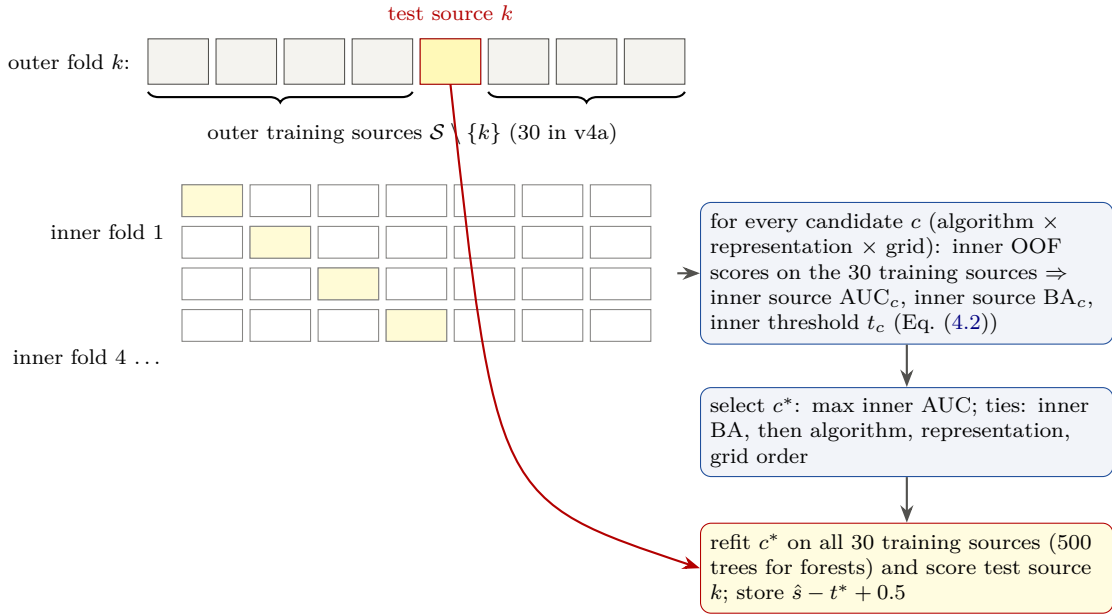


Figure 4.1: Nested LOSO as used in v4a (1d per algorithm and representation; 1e over all 140 configurations). For each outer fold, every candidate configuration is evaluated by an inner LOSO over the outer training sources only; the winner and its inner-optimal threshold are refitted on the outer training sources and applied once to the held-out source. The bias of “choosing the best of many” is therefore included in the outer estimate. Drawn for this report.

Other nested selections. In v7c2 (MAXI) the kNN k was selected by inner LOSO source AUC; the SVM (C, γ) by inner `GroupKFold(5)` source AUC (a pre-run deviation for computing time; ??); the first configuration with strictly larger AUC wins (`v7lib.c2_outer`). Every other model in the project uses fixed hyper-parameters, so no other selection exists.

4.5 Source bootstrap

The unit of generalisation is the source, so uncertainty is estimated by resampling sources, stratified by class, while keeping the OOF scores fixed (Figure 4.2). The draws are identi-

Algorithm 15 Nested automatic selection, v4a 1e (v4lib.nested_fold, 16_v4a_algorithms.py)

Input: outer folds; representations $\mathcal{R} = (\text{H}, \text{HI}, \text{B}, \text{A})$; algorithms $\mathcal{A}$ (8) with grids V4_GRID (35 settings)

```

1: for each outer fold  $k$  with training rows  $\mathcal{T}$  do
2:   for each  $(r, a) \in \mathcal{R} \times \mathcal{A}$  and grid entry  $c$  do
3:     inner LOSO over the sources of  $\mathcal{T}$  (stage “inner”: 200 trees, MLP seeds  $\{0, 1, 2\}$ )  $\rightarrow$ 
       inner source means
4:      $\text{AUC}_{rac} \leftarrow \text{Eq. (4.4)}$ ;  $(t_{rac}, \text{BA}_{rac}) \leftarrow \text{Eq. (4.2)}$ 
5:   end for
6:   per  $(r, a)$ :  $c^*(r, a) \leftarrow \arg \max (\text{AUC}, \text{BA}, -\text{grid index})$ ; refit (stage “outer”: 500 trees,
       seeds  $\{0..4\}$ ) and score fold  $k$   $\triangleright 1d$ 
7:    $(r, a)^* \leftarrow \arg \max (\text{AUC}, \text{BA}, -\text{algorithm order}, -\text{representation order}, -\text{grid index}) \triangleright$ 
       1e
8:   store  $\hat{s}_i - t^* + 0.5$  for  $i$  in fold  $k$ 
9: end for

```

cal in every version: BH and NS source names are sorted, `rng = numpy.random.default_rng(42)`, and draw b is $\mathcal{D}^{(b)} = (\text{rng.choice}(\mathcal{S}_{\text{BH}}, S_1), \text{rng.choice}(\mathcal{S}_{\text{NS}}, S_0))$, with replacement, $b = 1, \dots, 2000$ (09_bootstrap_sources.py, v3lib.bootstrap, v4lib.paired_bootstrap, v7lib.boot_draws). Every metric M is recomputed on the multiset $\mathcal{D}^{(b)}$ (a source drawn twice counts twice); for AUC this means Eq. (4.4) over the drawn BH and NS source scores. The 95% interval is the percentile interval

$$\text{CI}_{95}(M) = [Q_{0.025}\{M^{(b)}\}, Q_{0.975}\{M^{(b)}\}]. \quad (4.6)$$

Paired differences. Two models evaluated on the same sources use the same draws: $\Delta^{(b)} = M_A^{(b)} - M_B^{(b)}$, point estimate $M_A - M_B$ on the original sample, CI by Eq. (4.6) on $\{\Delta^{(b)}\}$, and $\text{frac_first_better} = \frac{1}{2000} \sum_b \mathbf{1}[\Delta^{(b)} > 0]$.

Decision rule used throughout (pre-registered from v4 on)

A difference is called **detected** only if its 95% bootstrap CI excludes 0. Otherwise it is **not detected under the pre-registered criterion** – never “no difference”, and an interval touching 0 (e.g. $[0.000, +0.179]$) is *not* detected. Each metric is judged separately; source BA and source AUC are the headline metrics.

What the interval does and does not contain. Models are not refitted inside the bootstrap, so the interval reflects which sources happen to be in the sample, not the variability of model fitting or model selection. Because sources of one class are treated as exchangeable, very heterogeneous NS sub-classes (Z sources, atolls, AMXPs) or unusual BHs (GRS 1915+105) can make the interval optimistic. With 7 BH sources the BH side of every interval is coarse: a single BH source changes the source BA by $1/(2 \cdot 7) \approx 0.07$, which is why many intervals have endpoints exactly at 0.

State-stratified variant (v7a, v8, v9a, v10a, v11a, v12b, v13a). For observation-level metrics inside a state (soft-like, hard-like, ...), sources are drawn from the whole sample (so the draws are the same as above), and the metric is computed on the drawn sources’ observations that belong to the state (v8lib.strat_boot). If a draw contains no BH or no NS observation in the state, the metric is NaN and is ignored in the percentiles (the number of valid draws is reported). Differences between strata (“soft – hard”) and between models are paired by draw.

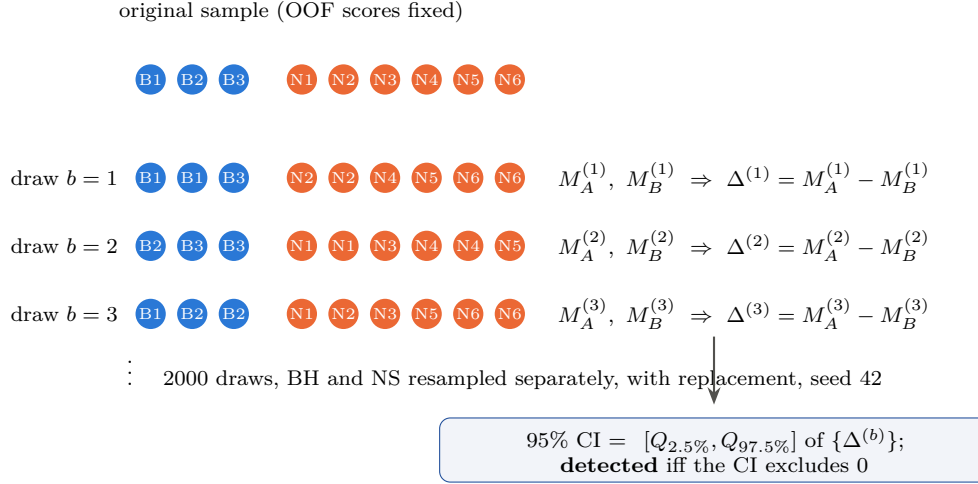


Figure 4.2: Class-stratified source bootstrap with paired differences. Each draw resamples the BH sources and the NS sources separately, with replacement and with all observations of a drawn source; both models are evaluated on the same drawn multiset, so the difference $\Delta^{(b)}$ is paired. No model is refitted. Drawn for this report.

Pooled variant (v10a). Sources of RXTE and NICER are mapped to one canonical name if they lie within 0.1° (`v10lib.canonical_map`); draws are taken over the union of sources; in each draw each component (RXTE v2, RXTE external, NICER) computes its own state-stratified AUC gain on its drawn sources, and the pooled statistic is the equal-weight mean; draws in which any component is undefined are dropped (`v10lib.pooled_boot`). A source present in two instruments is therefore always drawn (or not) in both.

4.6 Colour-conditional source permutation tests

Question. At matched colour, does a timing feature F (e.g. $\log_{10} \nu_c$) differ between BH and NS sources? **Statistic.** With the source mean residuals $\bar{r}_k$ of the label-free kNN regression (Eq. (3.15)):

$$D = \frac{1}{S_1} \sum_{k \in S_{\text{BH}}} \bar{r}_k - \frac{1}{S_0} \sum_{k \in S_{\text{NS}}} \bar{r}_k. \quad (4.7)$$

Null distribution. Under H_0 (given colour, F does not depend on the class) the source labels are exchangeable: $\Pi = 20\,000$ random relabellings keeping the number of BH sources fixed (`v6lib.perm_test`, seed 42), giving $D^{*(1)}, \dots, D^{*(\Pi)}$, and the two-sided p -value

$$p = \frac{1 + \#\{\pi : |D^{*(\pi)}| \geq |D| - 10^{-15}\}}{1 + \Pi}. \quad (4.8)$$

The source, not the observation, is permuted, so within-source correlation is respected. When several features are tested together (v6d: $T_1^*, T_2^*, T_3^*, \log_{10} \nu_c$), Holm's step-down adjustment is reported: for ordered $p_{(1)} \leq \dots \leq p_{(m)}$, $\tilde{p}_{(r)} = \max_{q \leq r} \min\{1, (m - q + 1)p_{(q)}\}$ (`v6lib.holm`). Decision: $p < 0.05$ (primary features); synthetic calibration: under H_0 the fraction of $p < 0.05$ was 0.033 over 60 data sets (`logs/v6/test_v6lib.log`).

Pooled permutation across instruments (v10b, also v12a pooled). Each instrument ι has its own residuals and D_ι ; the statistic is $D = \frac{1}{|I|} \sum_\iota D_\iota$. A source keeps one label in all instruments. The pre-registered null permuted labels over the union of sources; a synthetic test before any v10b statistic showed that sources present in only one instrument then absorb BH

Algorithm 16 Colour-conditional source permutation test

Input: observation colours $\mathbf{H}$, feature F , sources $\mathbf{g}$, source labels Y_k ; $K = 25$, $\Pi = 20000$

- ```

1: $\bar{r}_k \leftarrow$ Algorithm 13; keep sources with a finite $\bar{r}_k$; $D \leftarrow$ Eq. (4.7)
2: $\text{rng} \leftarrow \text{default_rng}(42)$
3: for $\pi = 1, \dots, \Pi$ do
4: assign BH to a random subset of S_1 sources; $D^{*(\pi)} \leftarrow$ Eq. (4.7)
5: end for
6: return D and p by Eq. (4.8)

```

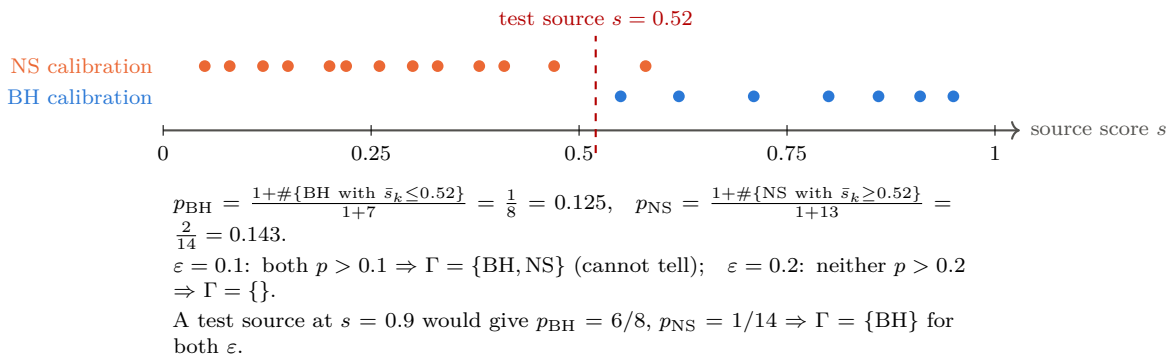
labels and widen the null, so labels are permuted *within strata of instrument membership* (RXTE only, NICER only, both), keeping the number of BH per stratum fixed (`v10lib.pooled_perm`). In synthetic data this gave a fraction 0.065 of  $p < 0.05$  under the null (200 data sets) and power 0.98 at a  $-1.5$  SD BH shift (`logs/v10/test_v10lib.csv`).

## 4.7 Source-level conformal prediction (v6e)

A conformal predictor returns, for a new source, a set of labels –  $\{\text{BH}\}$ ,  $\{\text{NS}\}$ ,  $\{\text{BH}, \text{NS}\}$  (“cannot tell”) or  $\{\}$  – with a finite-sample coverage guarantee under exchangeability. The class-conditional (Mondrian) version guarantees coverage within each class separately, which matters with 7 BHs. **Nonconformity** of a calibration source with score  $\bar{s}$ :  $\alpha = 1 - \bar{s}$  if BH,  $\alpha = \bar{s}$  if NS. For a test source with score  $s$  and calibration sources of class  $y$  ( $n_y$  of them),

$$p_{\text{BH}}(s) = \frac{1 + \#\{k \in \text{cal}_{\text{BH}} : 1 - \bar{s}_k \geq 1 - s\}}{1 + n_{\text{BH}}}, \quad p_{\text{NS}}(s) = \frac{1 + \#\{k \in \text{cal}_{\text{NS}} : \bar{s}_k \geq s\}}{1 + n_{\text{NS}}}, \quad \Gamma_\varepsilon(s) \quad (4.9)$$

(`v6lib.conformal_sets`). If the test source is exchangeable with the calibration sources of its class,  $\Pr(Y \in \Gamma_\epsilon) \geq 1 - \epsilon$  for that class. **LOSO version** (S1): each source is the test point and the other sources’ OOF scores are the calibration set (`conformal_loso`). **Split version** (external set E): the calibration set is the S1 LOSO OOF source scores of the same ( $H_R$ ) model and the test points are the frozen-model scores of the external sources. Reported: per-class coverage, mean set size, number of correct singletons and of ambiguous sets, at  $\epsilon \in \{0.1, 0.2\}$ .



**Figure 4.3:** Illustration (made-up calibration scores) of the class-conditional conformal  $p$ -values of Eq. (4.9). A test source is compared with the calibration sources of each class separately; a label enters the prediction set if the test source is not too atypical for that class. Drawn for this report.



## 4.8 Observation-budget curves (v6f)

How many pointings does a new source need? For  $k \in \{1, 2, 3, 5, 8, 13, 21\}$  and 500 draws (seed 42), each source’s score is the mean OOF score of  $\min(k, n_s)$  of its observations drawn without replacement; source BA (threshold 0.5) and source AUC (rank form of Eq. (4.4)) are computed for each draw and summarised by the median and the 2.5–97.5% range (`v6lib.budget_curves`; S3 OOF scores, so a source has up to 1329 observations to draw from).

## 4.9 Agreement, bridging and reliability statistics

- **Agreement** between two models (v3a, v4): Spearman correlation of their OOF scores over common observations, and the fraction of identical observation-level predictions (`v3lib.agreement`, `v4lib.agreement_vs`).
- **Bridge check** (v4b2):  $H_R$  may replace H only if, on the v2 sample,  $H_R$ -LR and H-LR have paired source-BA and source-AUC intervals containing 0 and OOF scores with Spearman  $\geq 0.90$ .
- **Split-half reliability** (v6, v8c, v10c): the feature is computed separately from odd and even rows/segments; the Spearman correlation  $\rho_{1/2}$  between halves is stepped up to the full length with Spearman–Brown,  $\rho_{\text{full}} = 2\rho_{1/2}/(1 + \rho_{1/2})$ .

## 4.10 Frozen and prospective protocol

Several tests were designed so that nothing about the model could adapt to the test data. Table 4.2 lists what was frozen and when. Two levels of independence must be distinguished: *source-level* independence (the test sources were never used in training: v6a, v8b, v11d) and *observation-level* independence (new observations of sources already used: v11, v13). Only the second was achievable for NICER BHs, because every dynamically confirmed BH with usable RXTE or NICER timing data had already been used.

## 4.11 Implementation checks and multiple comparisons

From v4 on, every version defined *implementation checks* (IC) that had to pass before any result was looked at: reproduction of earlier OOF predictions to  $\leq 10^{-8}$ – $10^{-15}$ , synthetic signals recovered by the timing estimators within 10%, pure-Poisson data giving  $|z| < 3$ , sanity checks with known sources (XTE J1118+480 hard-like, Z sources soft-like), and agreement with independent tools (XSPEC). A failed IC stops the sub-analysis or downgrades it to exploratory (v8c). Unit tests are in `scripts/test_v4lib.py`–`test_v11lib.py`.

**Multiplicity.** Each sub-analysis declares one primary comparison; all others are descriptive and are not used to claim a difference. No family-wise correction is applied across descriptive comparisons; where many are made (v4a: 76 combinations  $\times$  3 metrics = 228 intervals; v5: 45), the report states how many intervals would exclude 0 by chance ( $\approx 2.5\%$  one-sided per interval, but strongly correlated) and draws no conclusion from them. Within a permutation family (v6d) Holm adjustment is reported.



**Table 4.2:** What was frozen in the frozen and prospective tests, and when. “Out-of-source” = each source’s new observations are scored by the model fitted without that source.

| Test | Frozen before the test data were read                                                                                                                 | Model application                                                          | Record                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| v6a  | $H_R$ -LR, $H_R$ +T-LR, CCTLR( $H_R$ ) fitted once on S1; imputation medians from S1                                                                  | all external sources scored by the same S1 models                          | pre-reg. v6; models refitted deterministically in v8b (IC2: max diff $1.1 \times 10^{-16}$ ) |
| v8b  | the v6 models; v7a state rule                                                                                                                         | external set X (incl. GS 1354–64, SS 433)                                  | pre-reg. v8 ( <a href="#">a481fa0</a> )                                                      |
| v11  | NICER feature pipeline ( <code>v9lib.obs_features</code> ), state thresholds, models fitted on the v10 NICER full data, $\nu^* = 1.3007$ Hz           | out-of-source for the 60 v9 sources; all-data model for never-used sources | pre-reg. <a href="#">6fea016</a> ; $\nu^*$ <a href="#">58ded89</a>                           |
| v13  | v12 pipeline incl. 3C50 correction and screening (Eq. (1.21)), RF selected as primary, models fitted on the 534 v12 observations, $\nu^* = 1.3361$ Hz | out-of-source                                                              | pre-reg. <a href="#">42fe1d2</a> (with $\nu^*$ )                                             |

## Chapter 5

### Phase 1

# Bibliography